Population structure and admixture proportions

Ida Moltke

Outline for this session

- Introduction and motivation
  - What is population structure and admixture proportions?
  - What can we use it for?
- Methods for inferring population structure
  - ADMIXTURE
  - PCA (next session)
- Inferring admixture proportions from data
  - Practical issues and solutions

What is population structure?

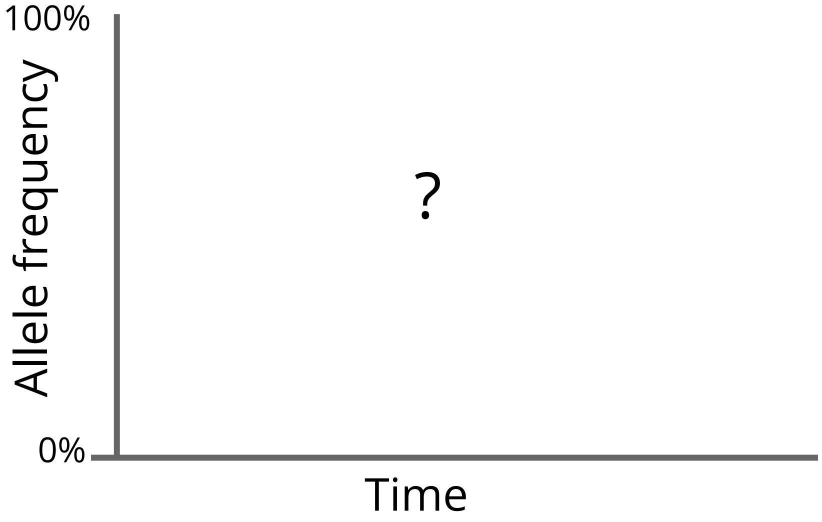
Wikipedia definition:
Population structure is the presence of a systematic difference in allele frequencies between subpopulations in a population possible due to different ancestry

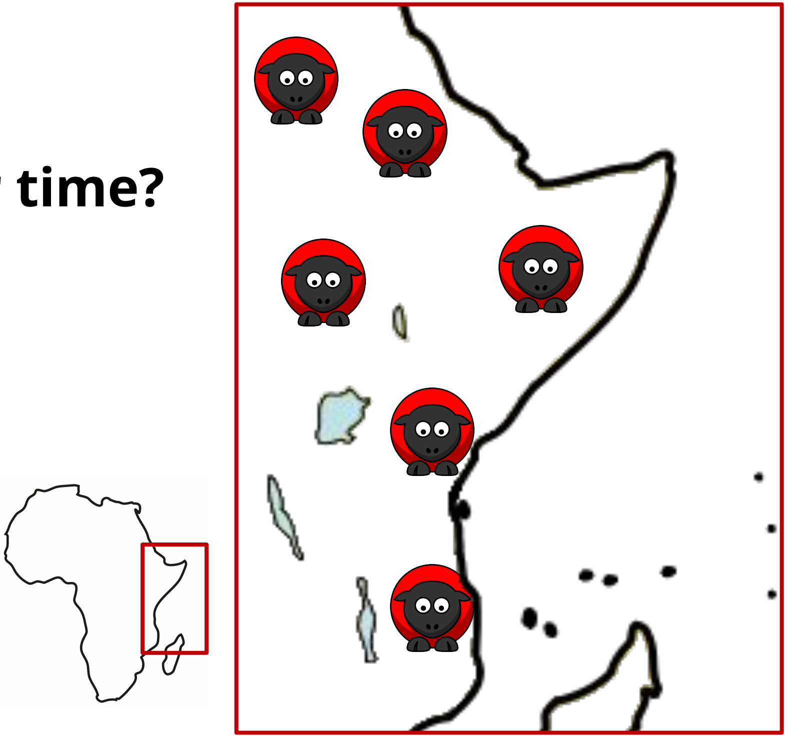
- (cur | prev) ○ 07:30, 17 June 2008 Andersduck (talk | contribs) . . (7,245 bytes) **(+7,245)** . . (← *Created page with 'Population stratification is the presence of a systematic difference in allele frequencies between subpopulations in a population possible due to different ancestry...'*)

No population structure with random mating

If we look at a random SNP:

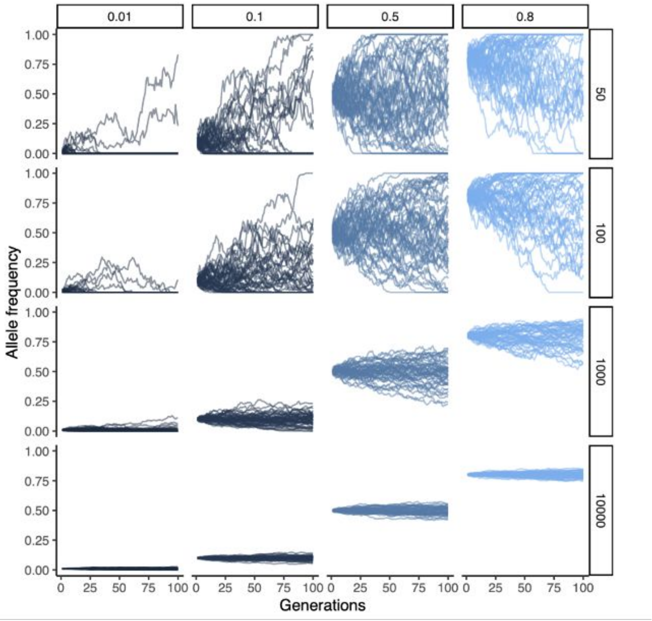
What happens to the frequency over time?





No population structure with random mating

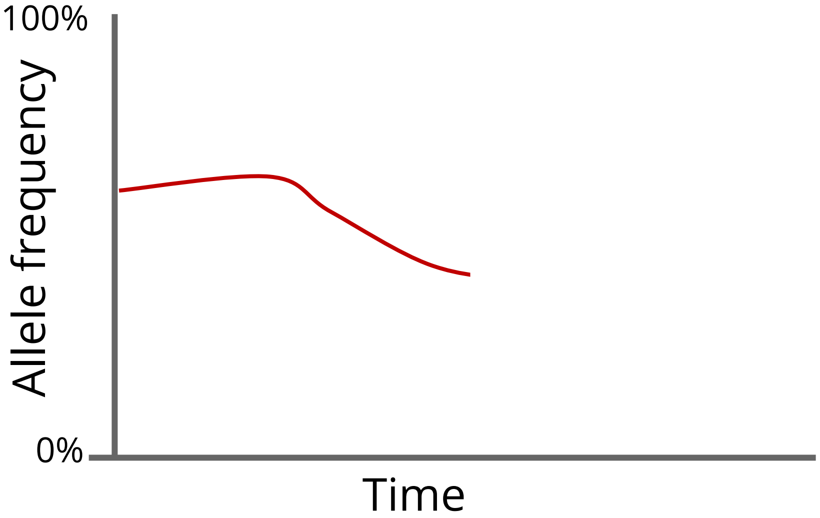
This is what we learned in Harvi's class:

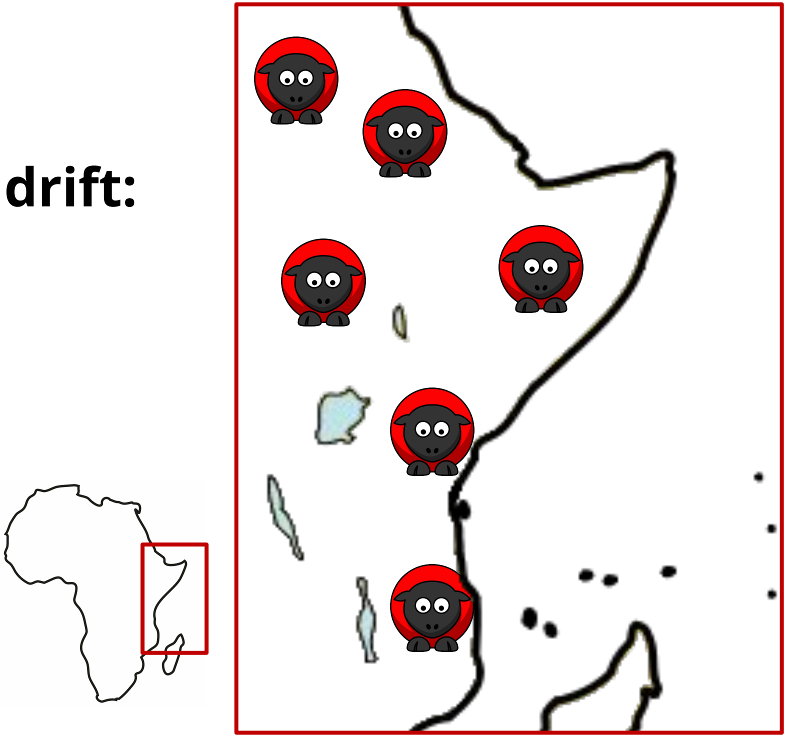


No population structure with random mating

If we look at a random SNP:

Frequency changes over time due to drift:

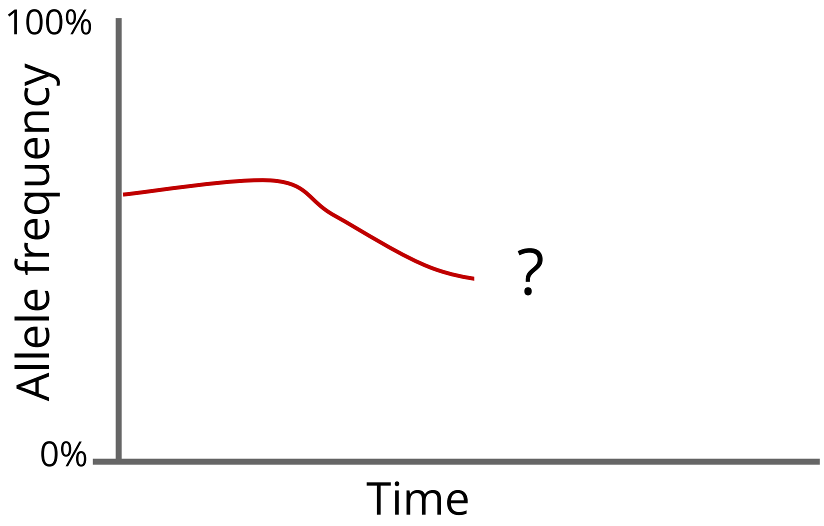


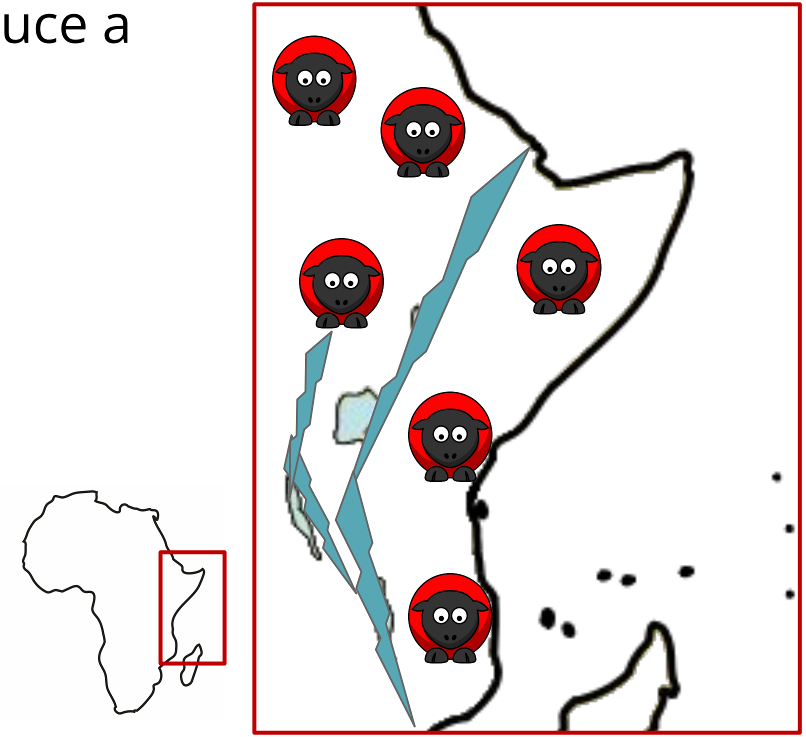


Barriers to gene flow leads to population structure

If we look at a random SNP **and** introduce a barrier to gene flow (e.g. a large river):

What will happen over time?

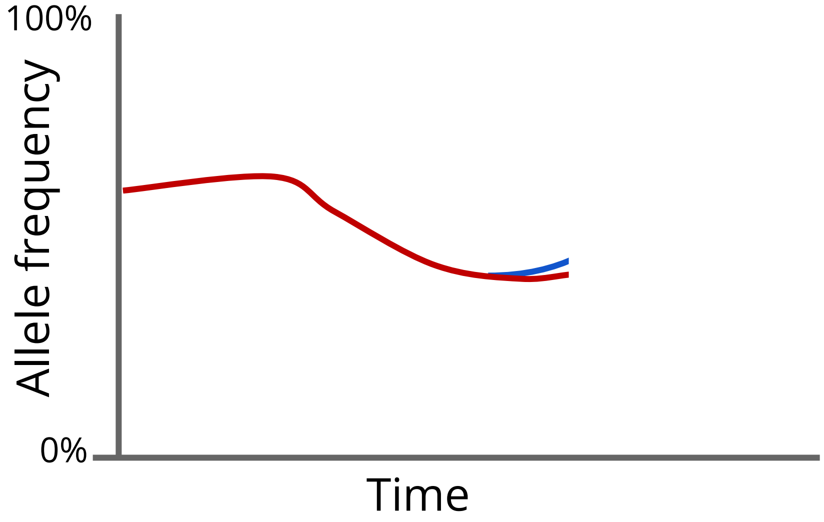


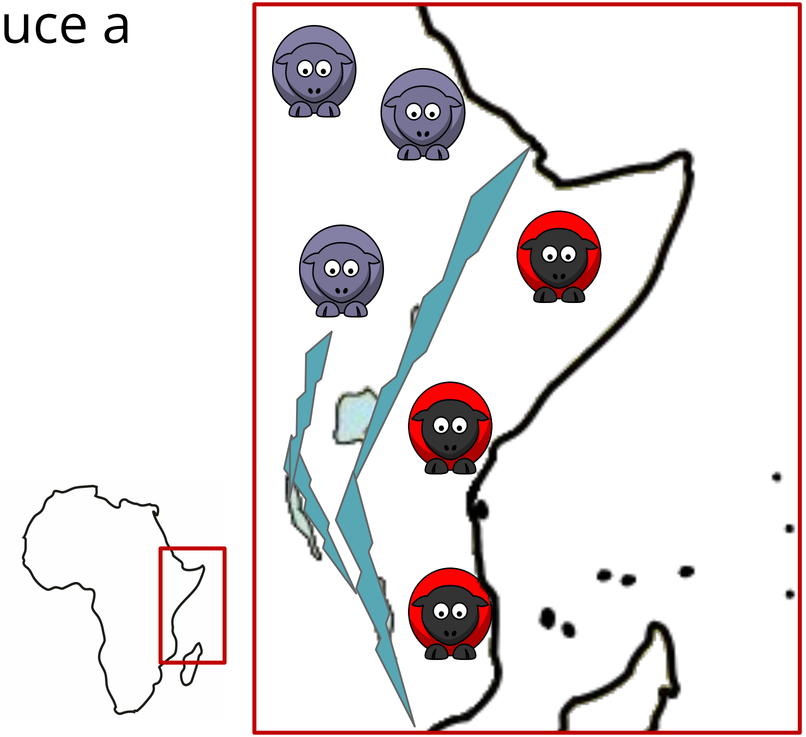


Barriers to gene flow leads to population structure

If we look at a random SNP **and** introduce a barrier to gene flow (e.g. a large river):

Frequency changes independently:

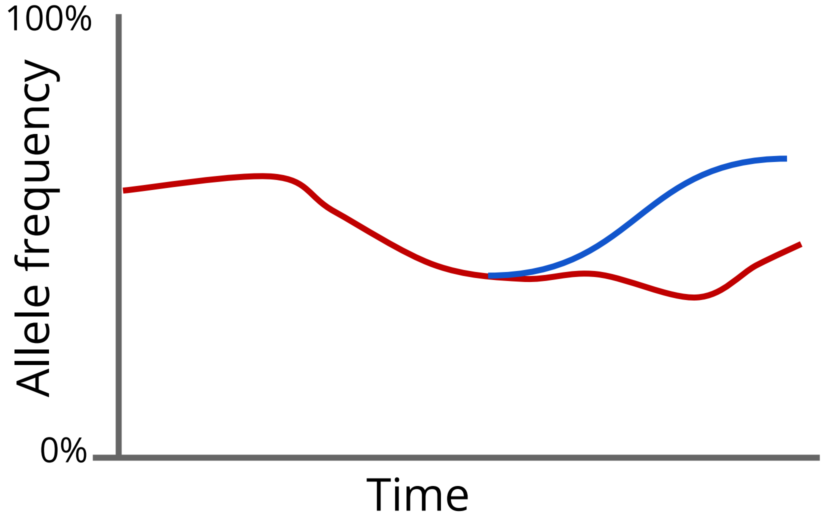


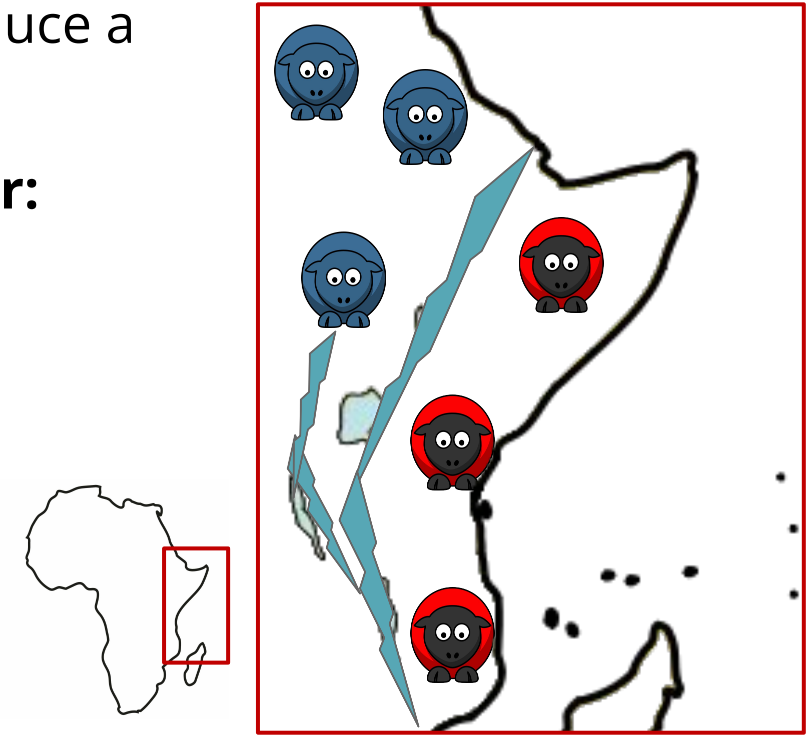


Barriers to gene flow leads to population structure

If we look at a random SNP **and** introduce a barrier to gene flow (e.g. a large river):

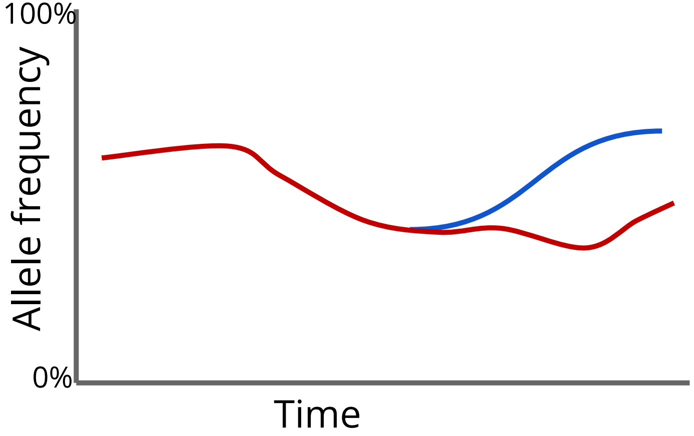
And over time frequencies will differ:

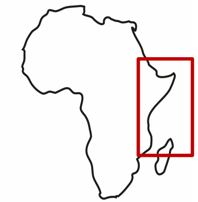


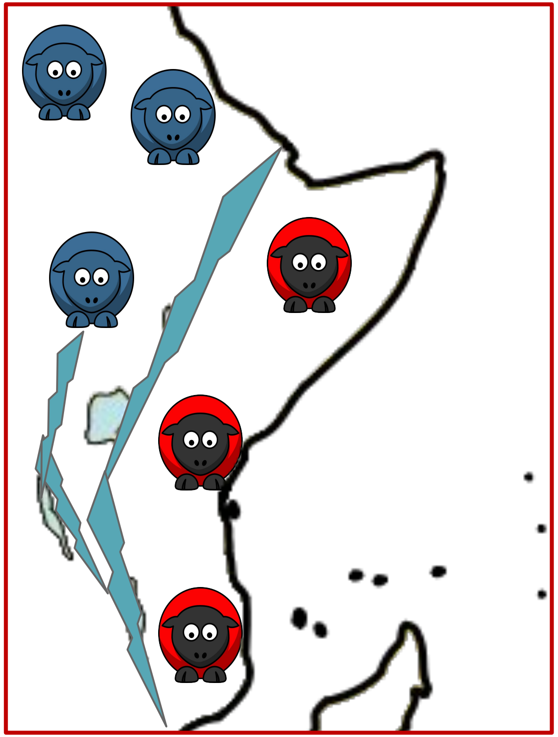


Affects all genetic variants!

SNP1







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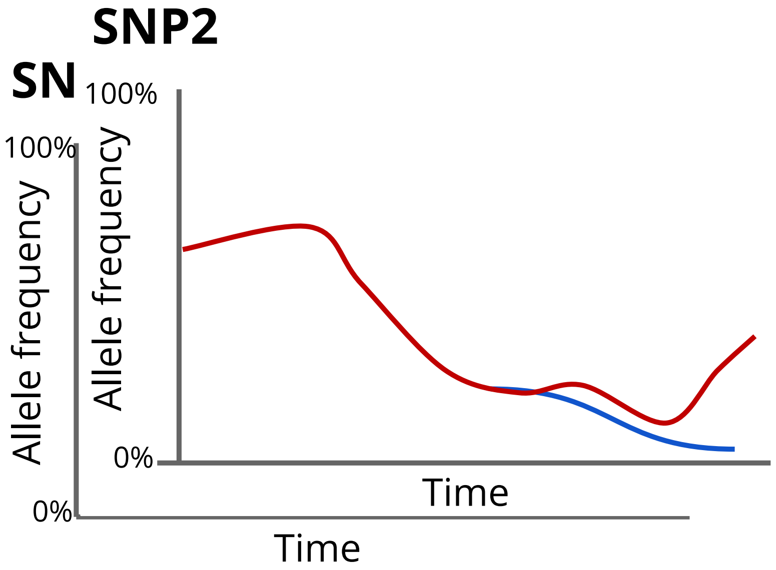

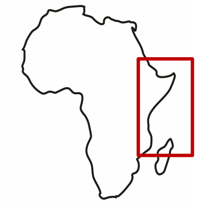

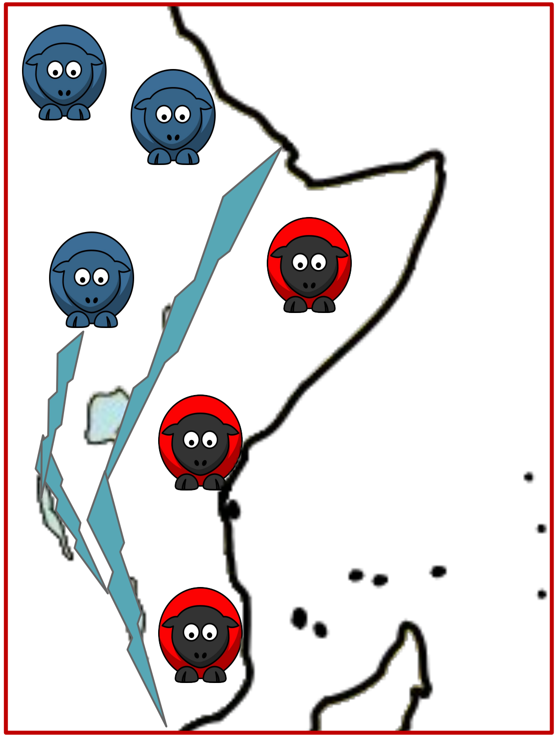

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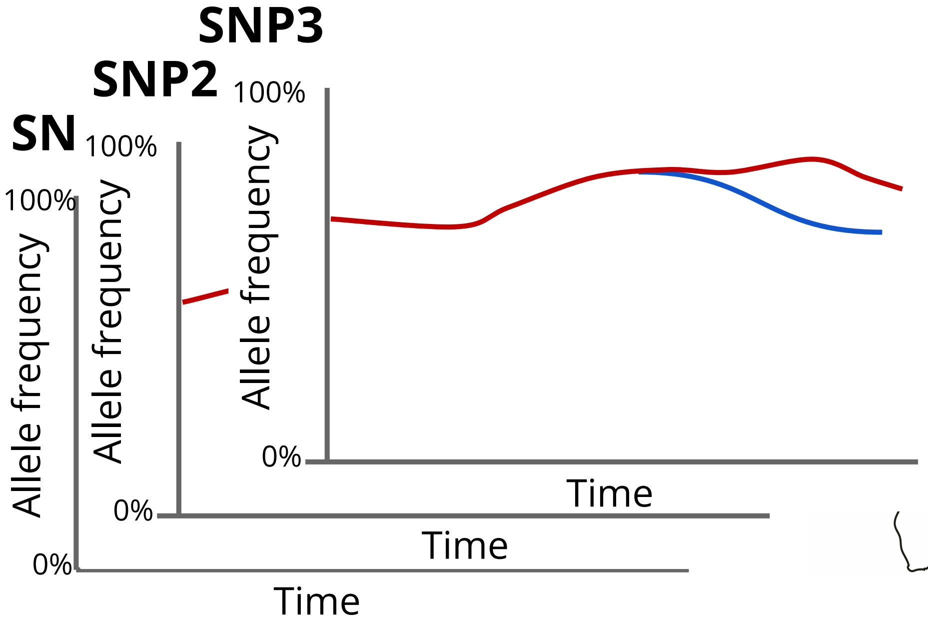
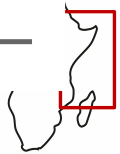
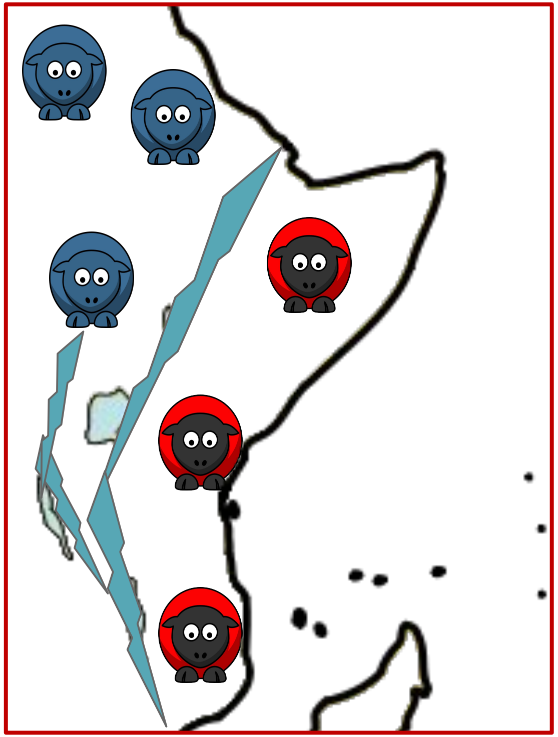
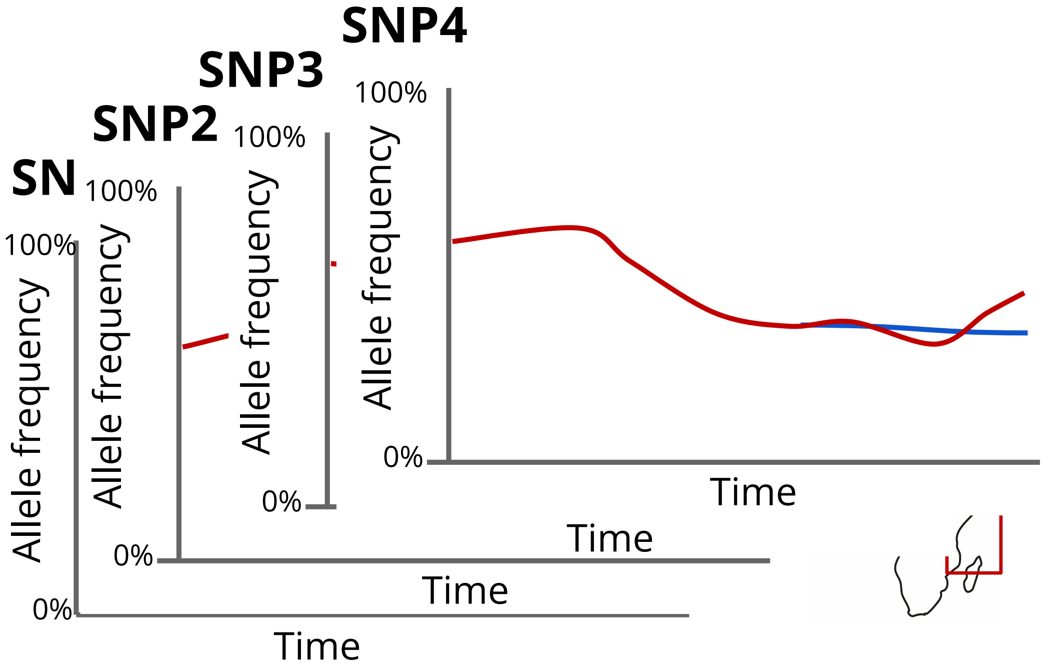
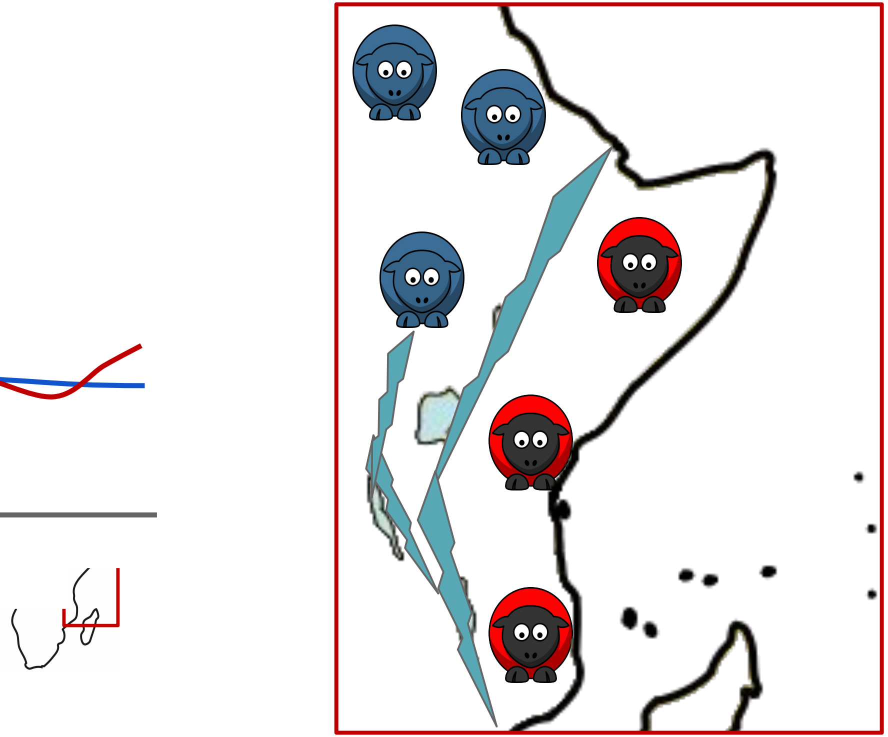






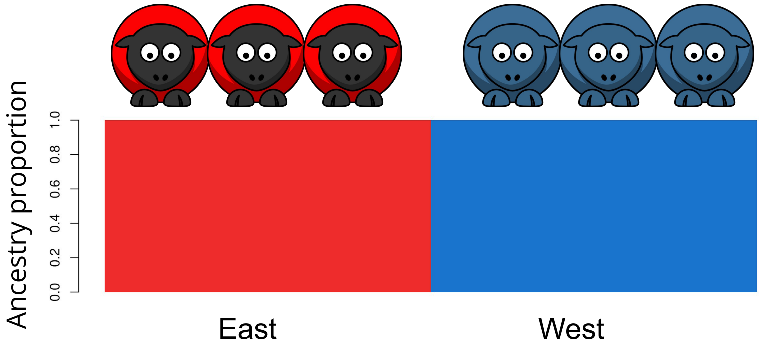
Affects all genetic variants!

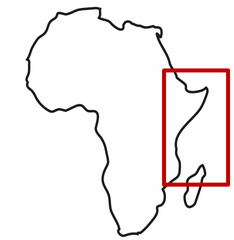


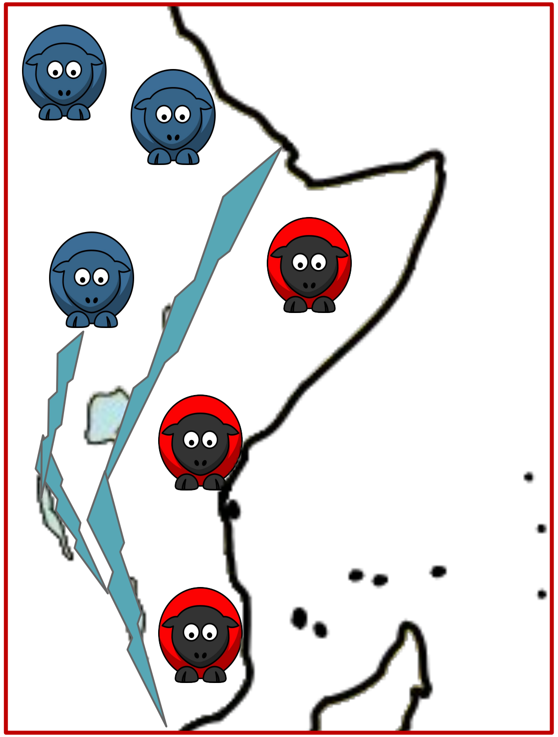


Barriers to gene flow leads to population structure

This **systematic allele frequency** difference allows us to separate individuals from the two subpopulations based on their DNA







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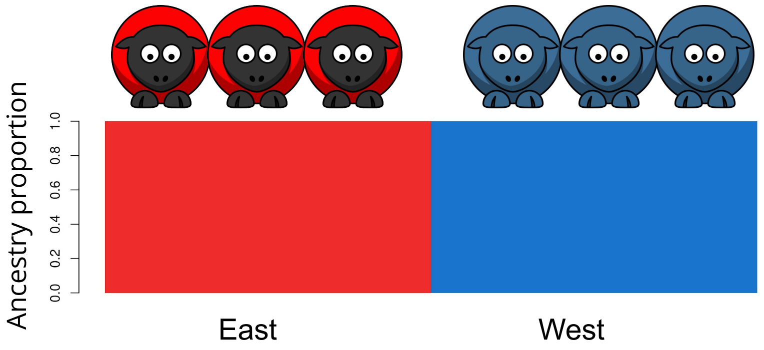
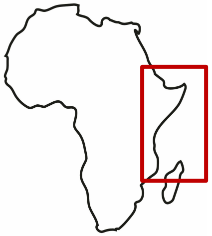
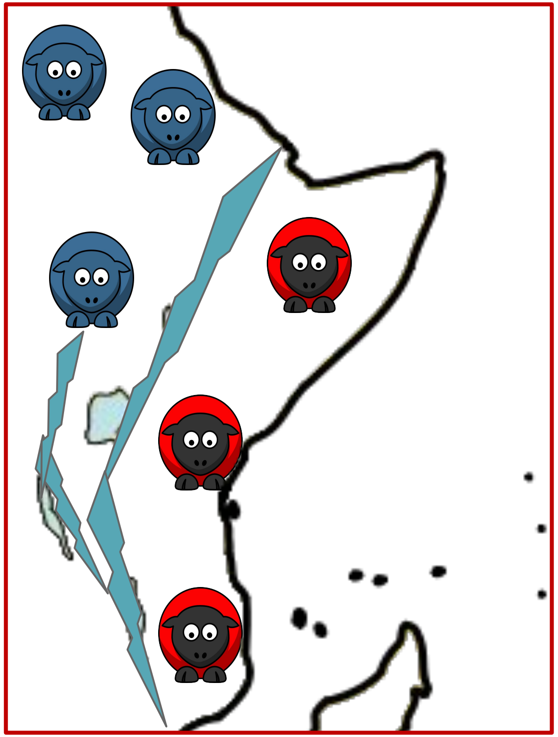
This is an example of population structure







What is population structure?

Hopefully that example makes Anders' definition a bit clearer:

Wikipedia definition:
*Population structure is the presence of a **systematic difference in allele frequencies between subpopulations** in a population possible due to different ancestry*

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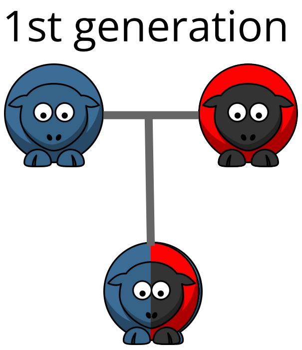
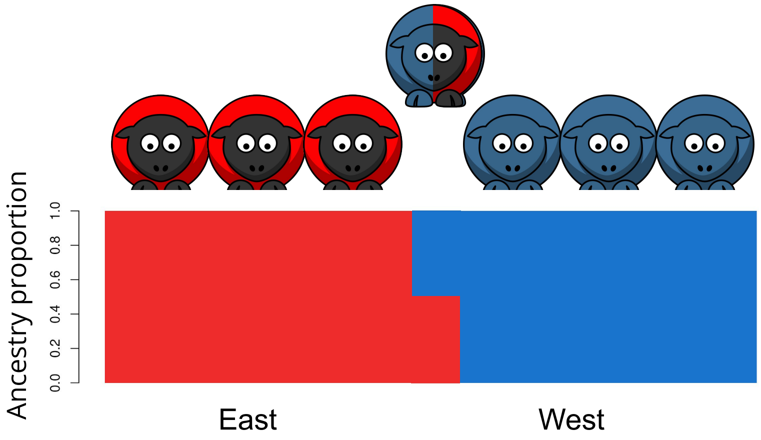
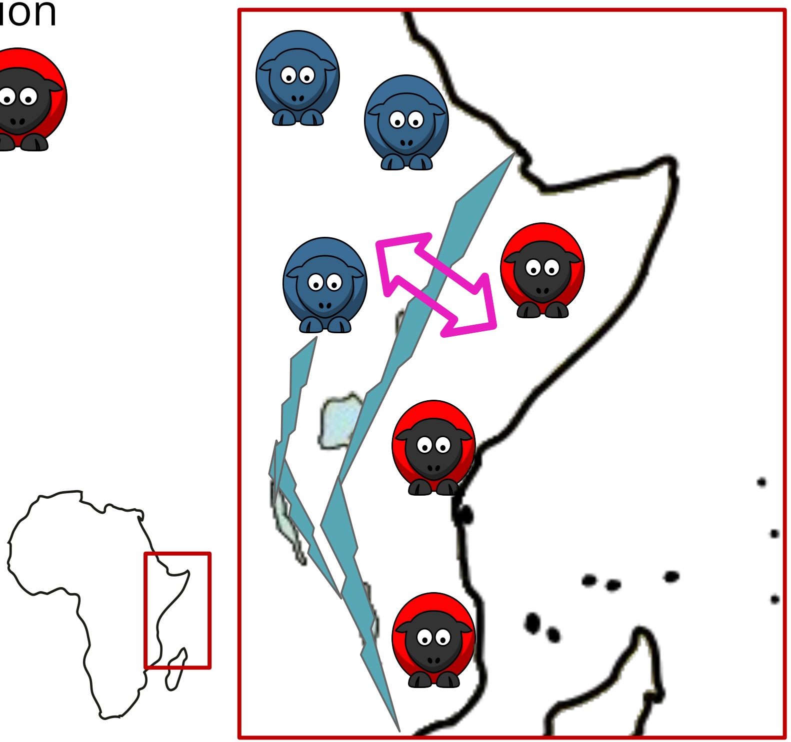
Another key term: Admixture (gene flow after separation)

We call individuals **admixed** if they have ancestry from several populations

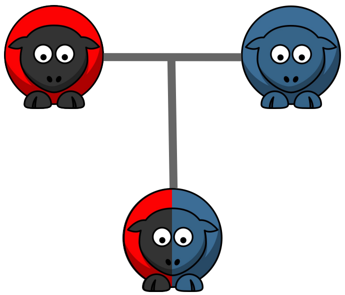


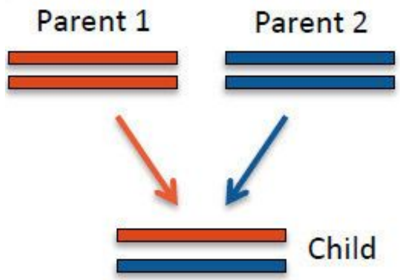




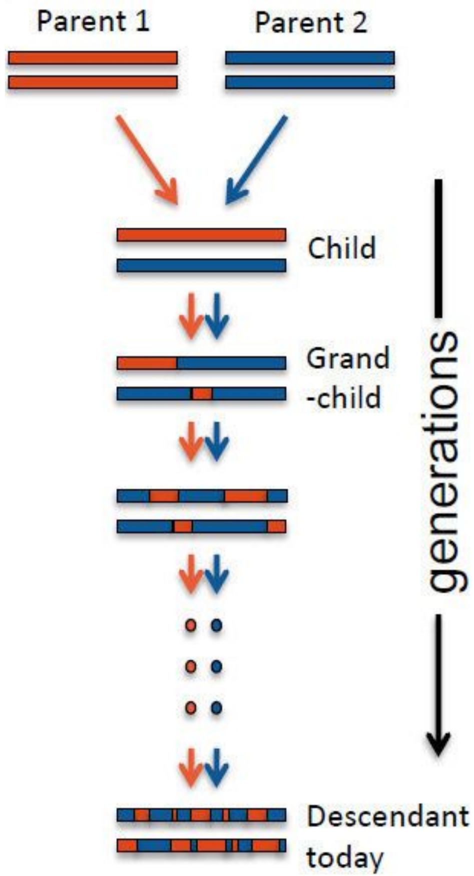
Admixture - at chromosome level

1st generation



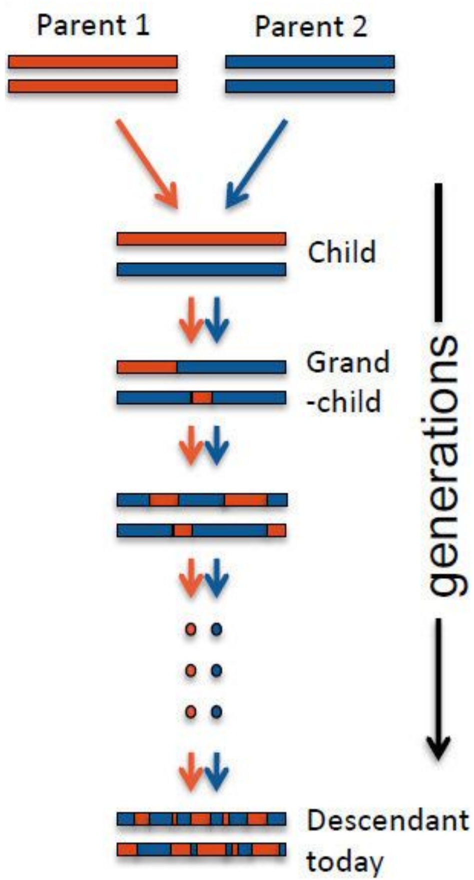


Admixture - at chromosome level



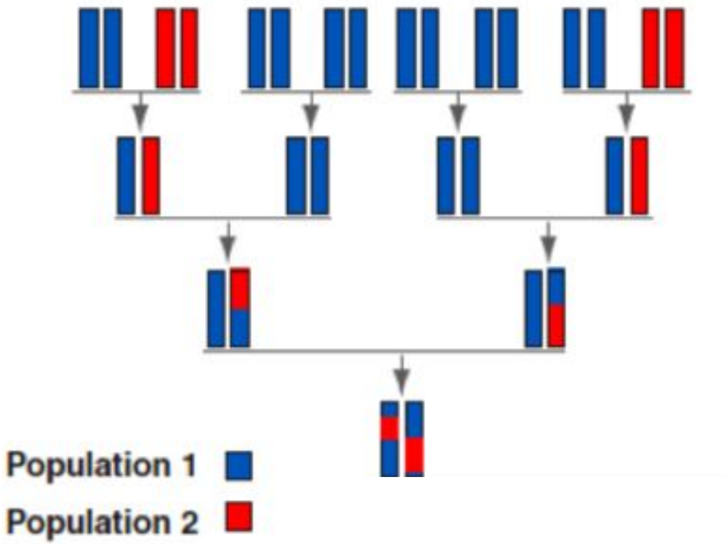
Over time the ancestry tracts will become shorter and shorter

Admixture - at chromosome level



Is it clear why it ancestry tracts will become shorter and shorter?

Illustrated in a pedigree

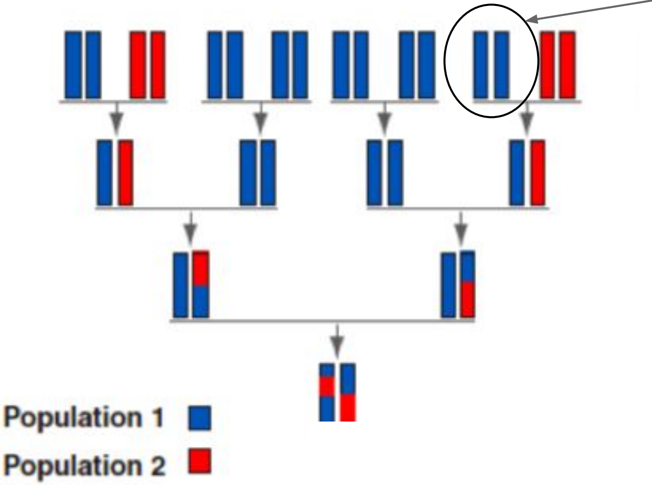


Key summary: ancestry proportions

A key summary statistic is ancestry proportions (or admixture proportion):
The proportion of a genome that originates from each (ancestral) population

Examples:

100% **population 1 (proportion = 1)**
0% **population 2 (proportion = 0)**



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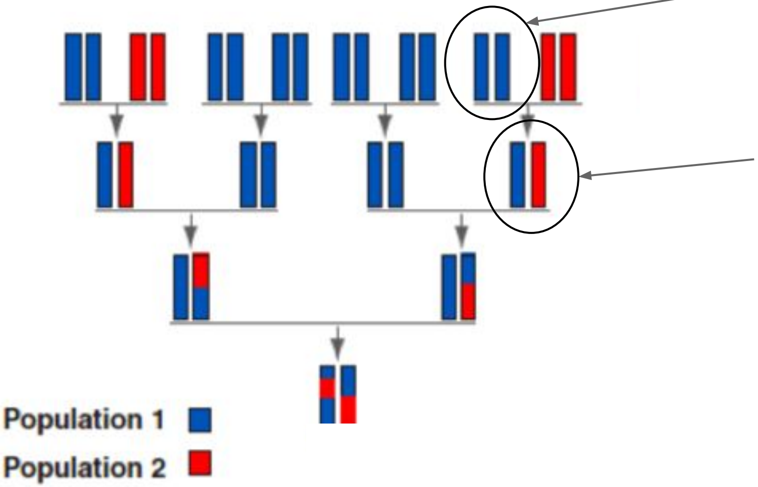

100% **population 1 (proportion = 1)**
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50% **population 1 (proportion = 0.5)**
50% **population 2 (proportion = 0.5)**

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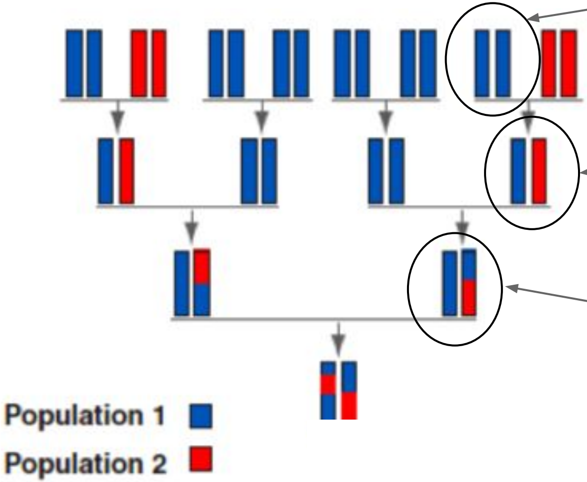

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?

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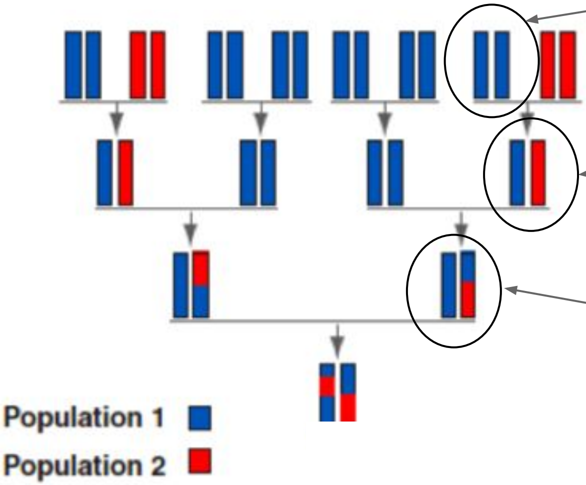
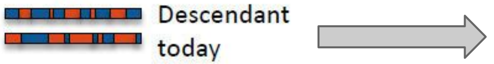


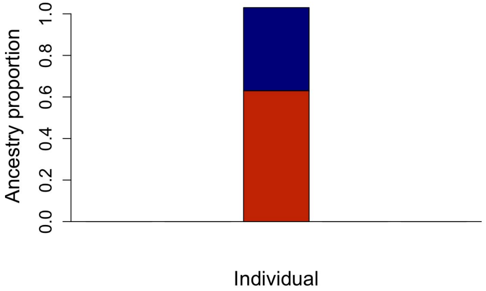
100% **population 1 (proportion = 1)**
0% **population 2 (proportion = 0)**

50% **population 1 (proportion = 0.5)**
50% **population 2 (proportion = 0.5)**

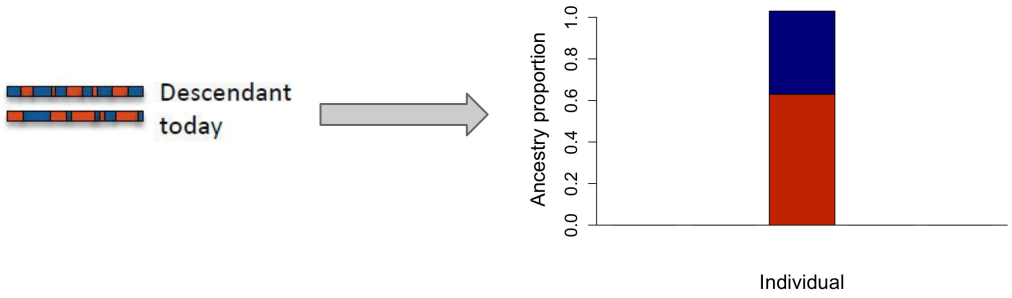
~80% **population 1 (proportion = 0.80)**
~20% **population 2 (proportion = 0.20)**

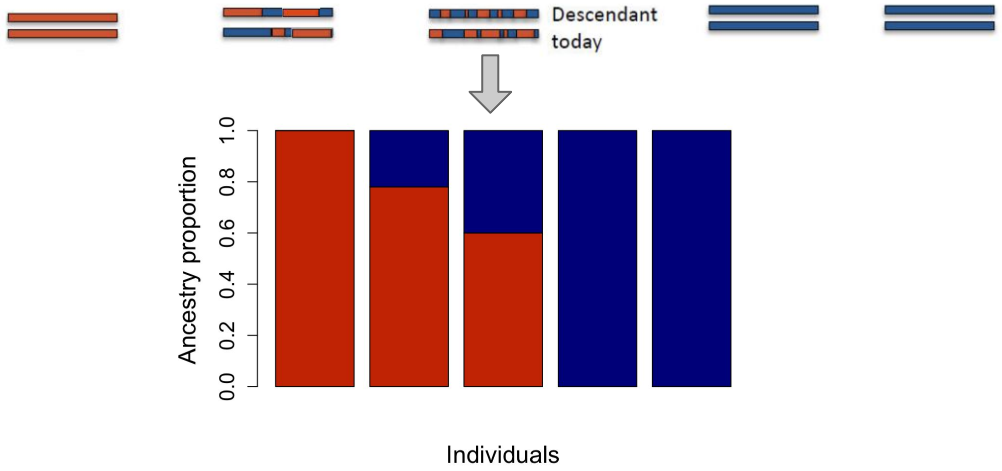
Typical plot: admixture plot



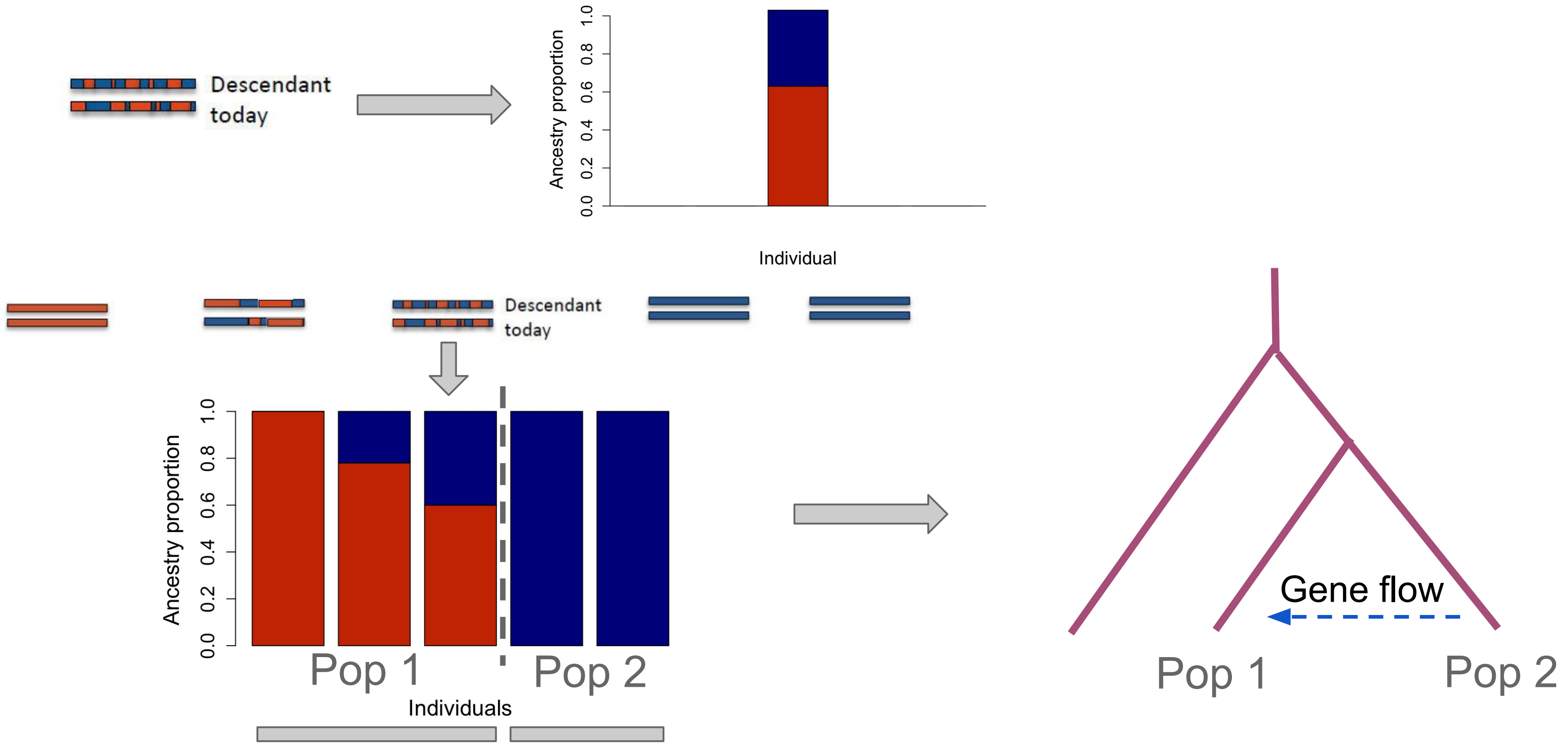


Typical plot: admixture plot

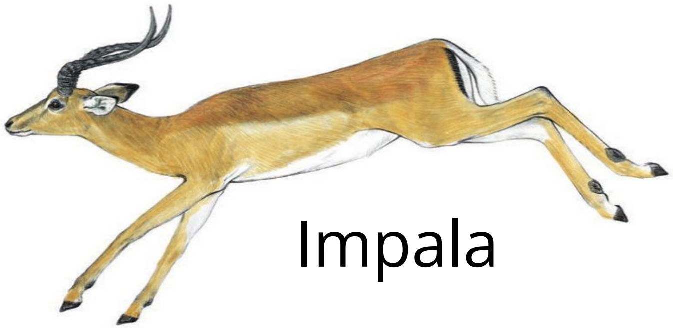


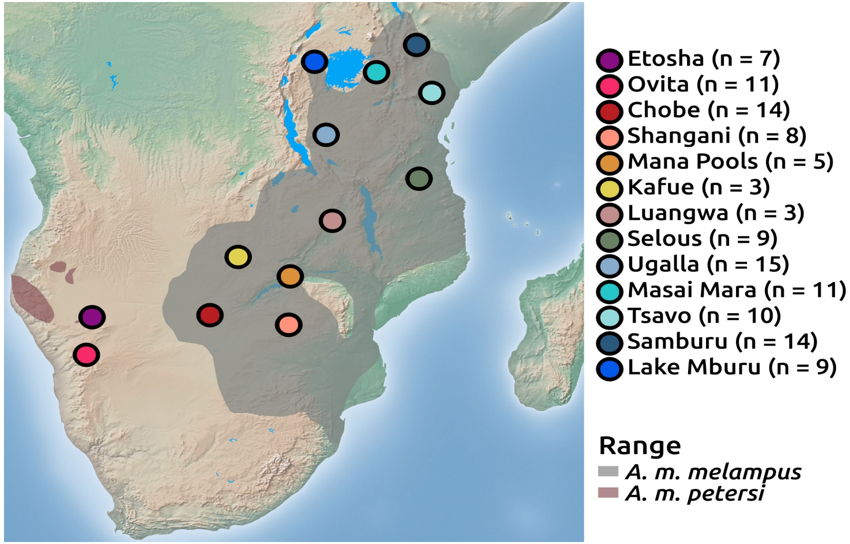


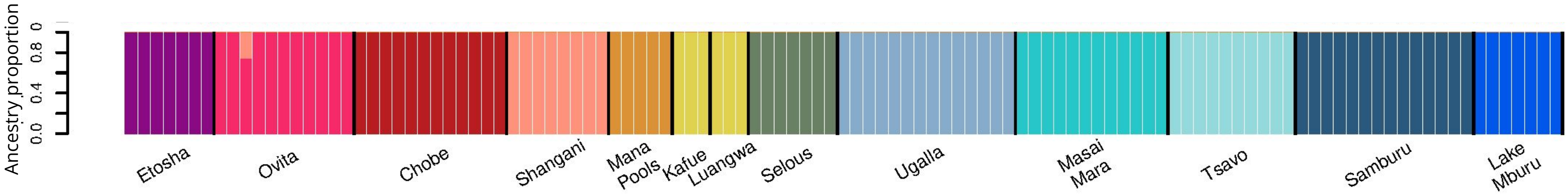
Can e.g. be used for demographic inference



Example: impala



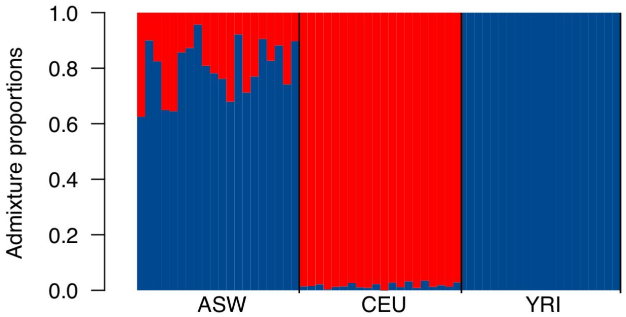




How to estimate ancestry proportions?

We want to infer ancestry proportions, Q (N×K matrix):

K=2



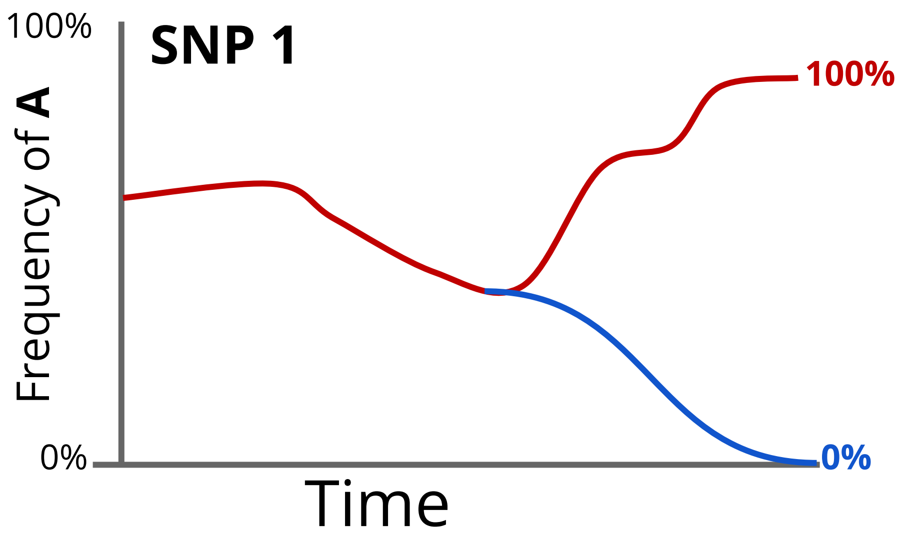
$$Q = \begin{array}{cl} \begin{pmatrix} 0.62 & 0.38 \\ 0.90 & 0.10 \\ 0.86 & 0.14 \\ \multicolumn{2}{c}{\cdots} \\ 0.00 & 1.00 \\ 0.00 & 1.00 \\ 0.00 & 1.00 \\ \multicolumn{2}{c}{\cdots} \\ 1.00 & 0.00 \end{pmatrix} & \begin{array}{l} \text{Ind\_1} \\ \text{Ind\_2} \\ \text{Ind\_3} \\ \cdots \\ \text{Ind\_21} \\ \text{Ind\_22} \\ \text{Ind\_23} \\ \cdots \\ \text{Ind\_n} \end{array} \end{array}$$

But all we have is genotypes, G (N x M matrix):

$$G = \begin{array}{cl} \begin{array}{cccc} \text{SNP1} & \text{SNP2} & \ldots & \text{SNPM} \end{array} & \\ \begin{pmatrix} AT & GG & & AT \\ TT & CG & & AA \\ AA & CG & \cdots & TT \\ \cdots & \cdots & & \cdots \\ AT & GG & & AT \\ TT & CC & & AA \\ AT & CG & \cdots & TT \\ \cdots & \cdots & & \cdots \\ TT & CG & & AA \end{pmatrix} & \begin{array}{l} \text{Ind\_1} \\ \text{Ind\_2} \\ \text{Ind\_3} \\ \cdots \\ \text{Ind\_21} \\ \text{Ind\_22} \\ \text{Ind\_23} \\ \cdots \\ \text{Ind\_n} \end{array} \end{array}$$

Solution: ADMIXTURE

We exploit that allele frequencies differs



We genotype an individual

SNP 1 Genotype = AA

What can we say?

Solution: ADMIXTURE

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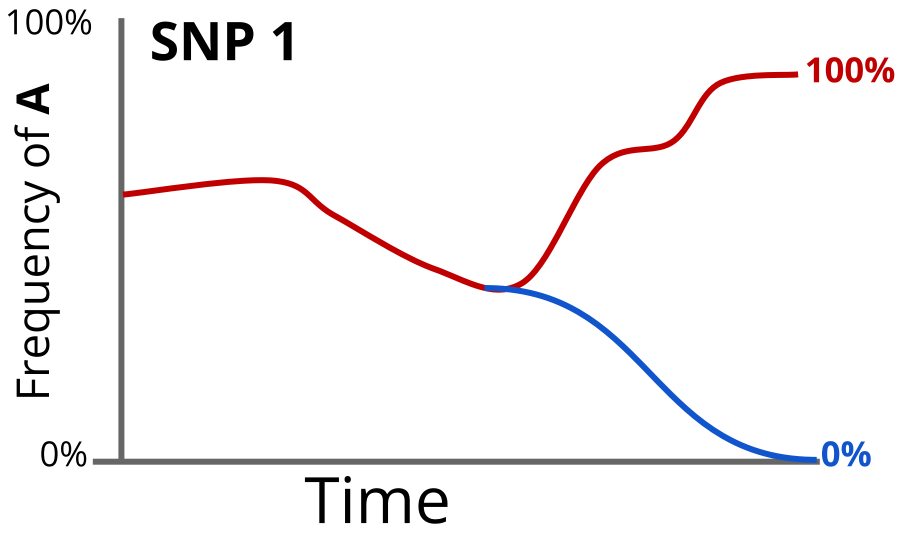

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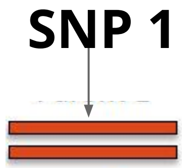

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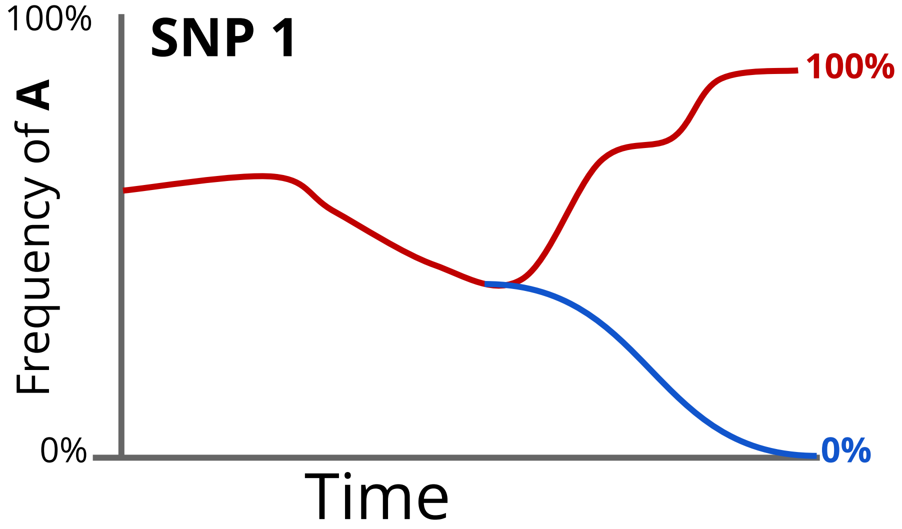

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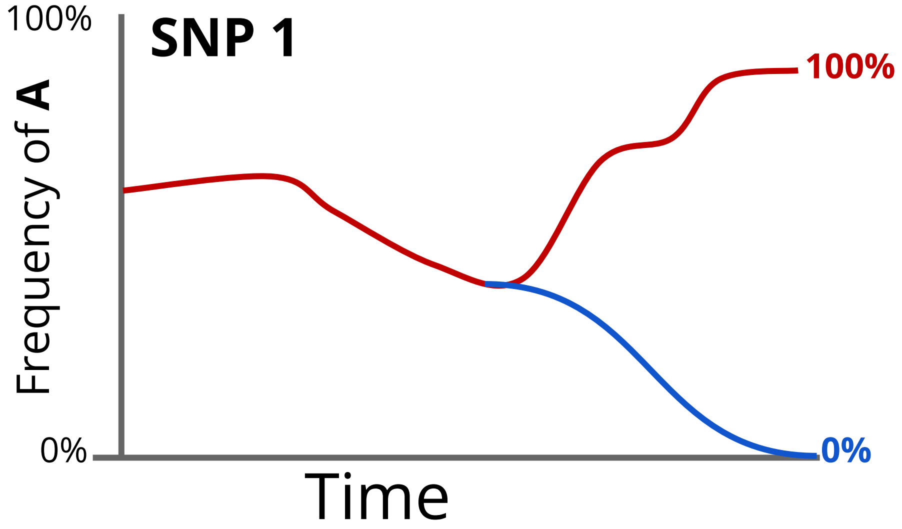

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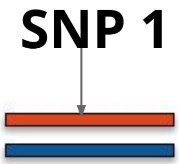

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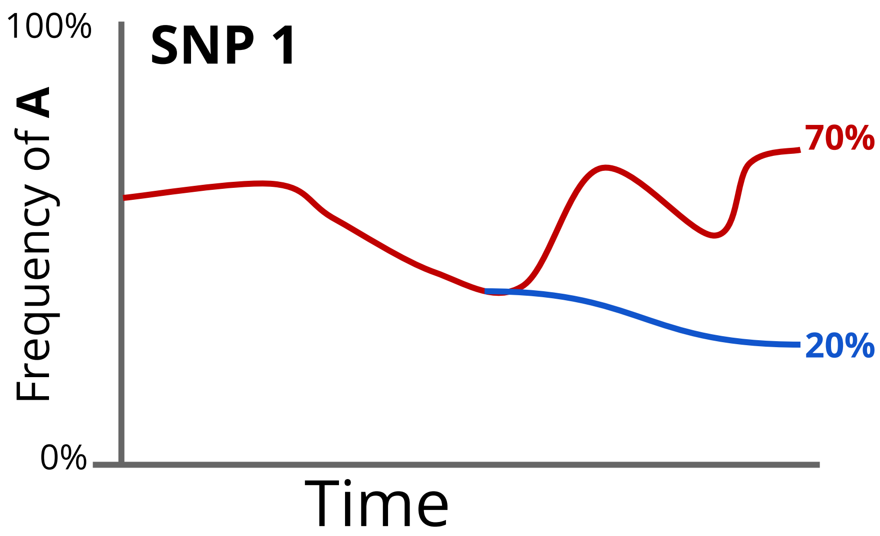


We genotype an individual

SNP 1 Genotype =AA

What can we say?

??

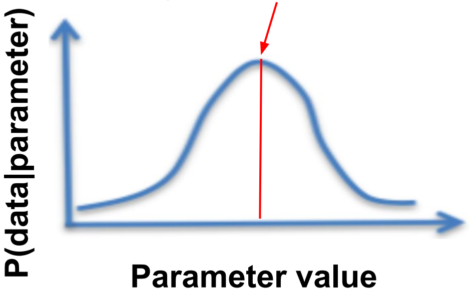
Solution: ADMIXTURE

- A model with genotypes (G) **across all SNPs and all individuals** as the observed data and ancestry proportions (Q) and ancestry-specific allele frequencies (F) as parameters
- The model consists of a formula for P(G|F,Q) (the likelihood):

$$P(G|Q,F) = \prod_i^N \prod_j^M p(G_{ij}|Q_i, F_j)$$

- It returns the F and Q that maximises that likelihood):

$$\hat{F}, \hat{Q} = argmax_{F,Q} P(G|F,Q)$$



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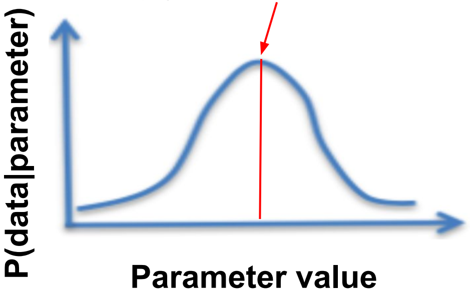
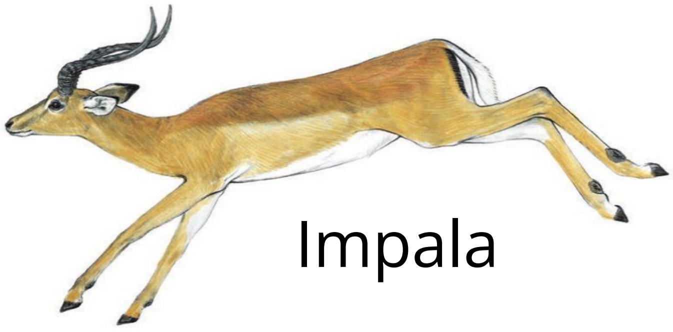
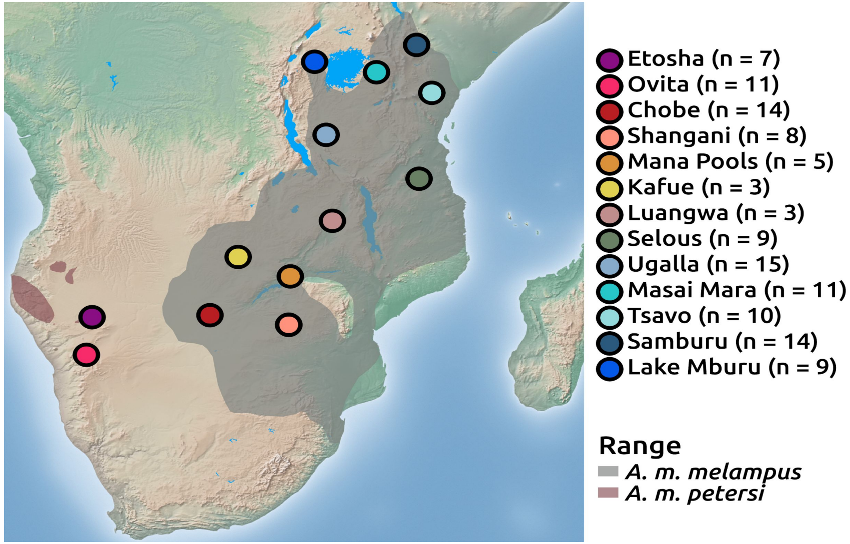
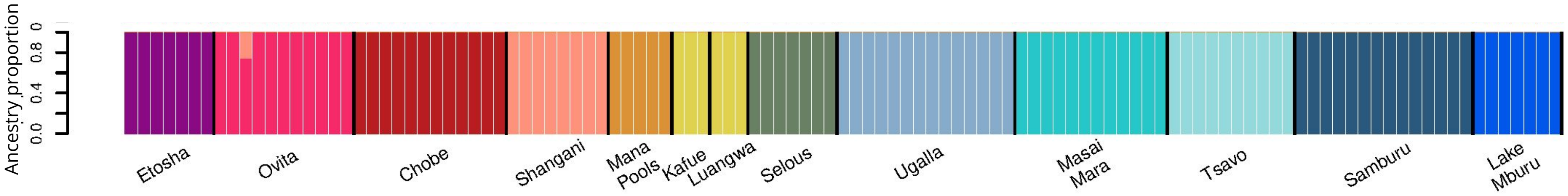


ADMIXTURE provides **the parameter values that make the observed data the most likely** to be observed

Works and produces results like this :-)

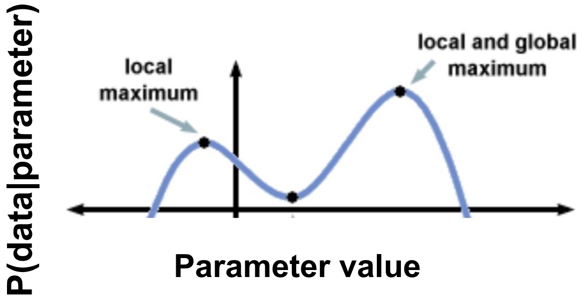






Issues to be aware of with ADMIXTURE - issue 1

It uses numerical optimization, because there is not an analytical solution!



So you need to run it several times to be able to assess whether it has found the maximum likelihood estimate (we say it “reached convergence”)!

Solution to issue 1: Assessing convergence

1. Run it ~10 times and sort them according to their likelihood:

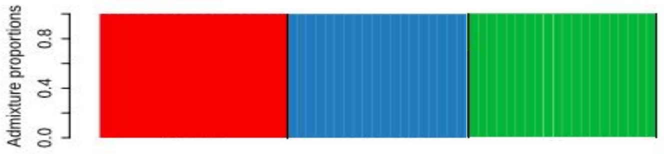
```
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles15K3.log -3865964.394787
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles18K3.log -3865964.414354
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles12K3.log -3865964.414777
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles5K3.log -3865964.415840
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles3K3.log -3865964.415979
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles7K3.log -3865964.419814
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles6K3.log -3865964.422308
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles13K3.log -3865964.424335
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles10K3.log -3865964.427415
/home/ida/teaching/popgen19/admixexercise/NGSadmixoutput/myoutfiles9K3.log -3865964.428677
```

2. If the top 5 have very similar likelihood pick the top one
3. If not, then run it 10 more times and go to step 2 again

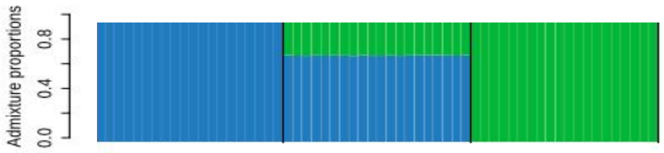
Issues to be aware of with ADMIXTURE - issue 2

Choosing K can be challenging. If K is too small you risk getting wrong results e.g.

Truth is K=3

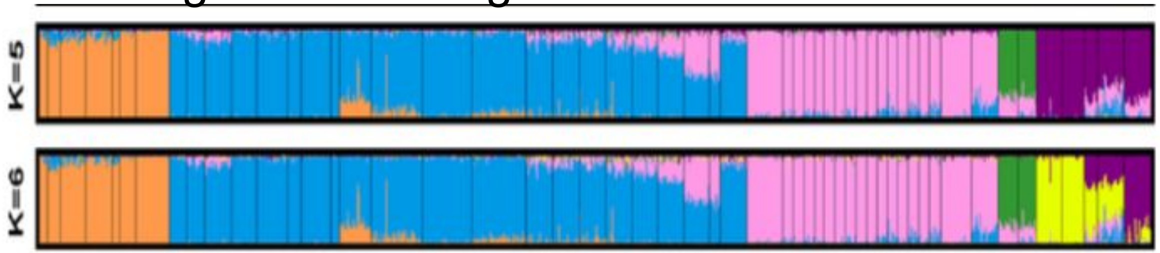


You choose K=2:



ADMIXTURE manual suggests to use cross-validation to find the “correct K”, but this approach has several issues, so I would never use that.

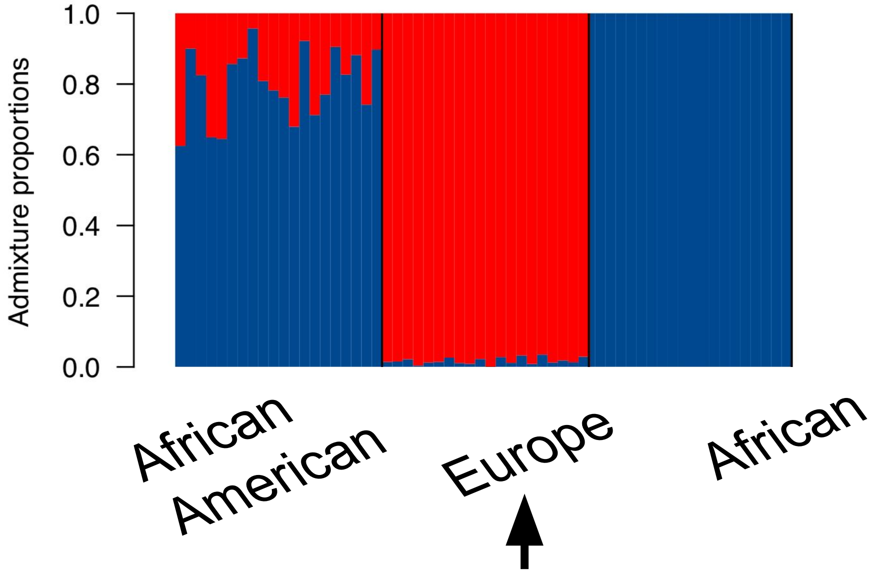
Also, several Ks might be meaningful…



Issues to be aware of with ADMIXTURE - issue 3

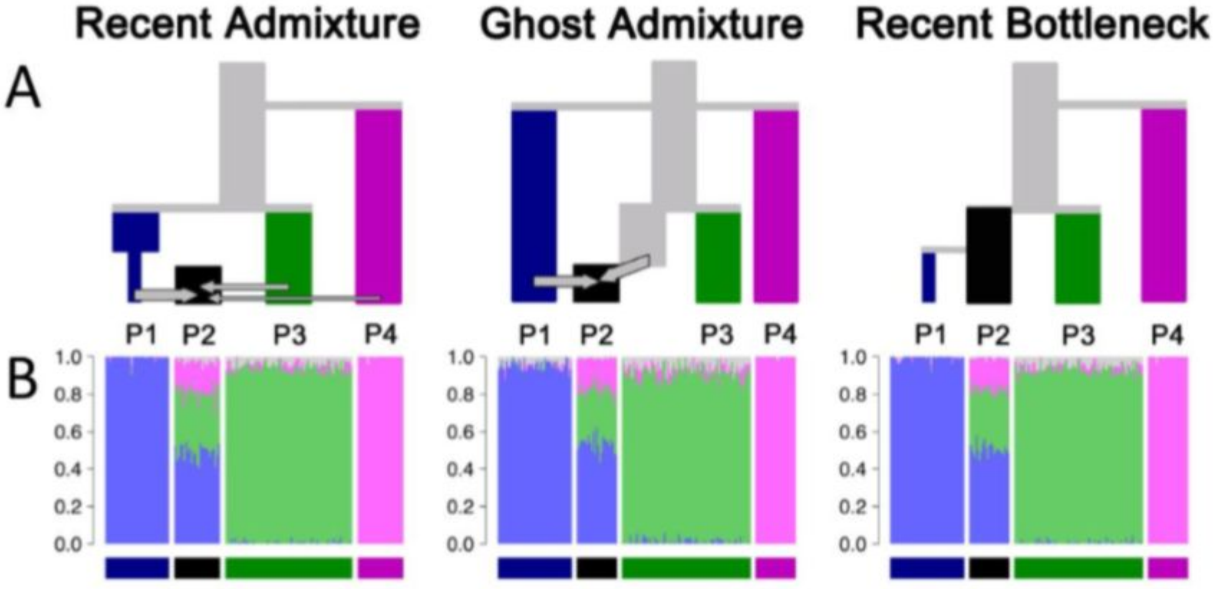
You may not have the real source populations and that can affect both the estimates and the interpretation

E.g. what if we had samples from China and not Europe?



Issues to be aware of with ADMIXTURE - issue 4

Several demographic scenarios can lead to similar results!



"A tutorial on how not to over-interpret STRUCTURE and ADMIXTURE bar plots" by Lawson et al. 2018

Potential solution to issues 2-4: evalAdmix

- There is a tool, **evalAdmix**, that can help assess the fit and suggest a lower bound for K
- For each individual i and locus j it looks at the difference between the observed genotype and the ADMIXTURE based estimate

$$r_{ij} = g_{ij} - \hat{g}_{ij} \qquad \hat{g}_{ij} = 2\sum_{k=1}^{K} q_{ik} f_{jk}$$

- More specifically it calculates the mean correlation between these differences for each pair of individual a and b

$$E[\hat{\rho}_{ab}]$$

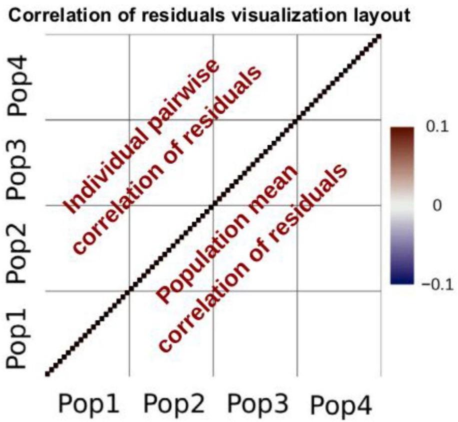
Potential solution to issues 2-4: evalAdmix

If individual *a* and *b* are sampled from the same population, and they have a good model fit, all error will be random and residuals will be uncorrelated:

$$E[\hat{\rho}_{ab}] = 0$$

If individual *a* and *b* are sampled from the same population AND they have a bad model fit, they will share a systematic error and their residuals will be positively correlated:

$$E[\hat{\rho}_{ab}] > 0$$



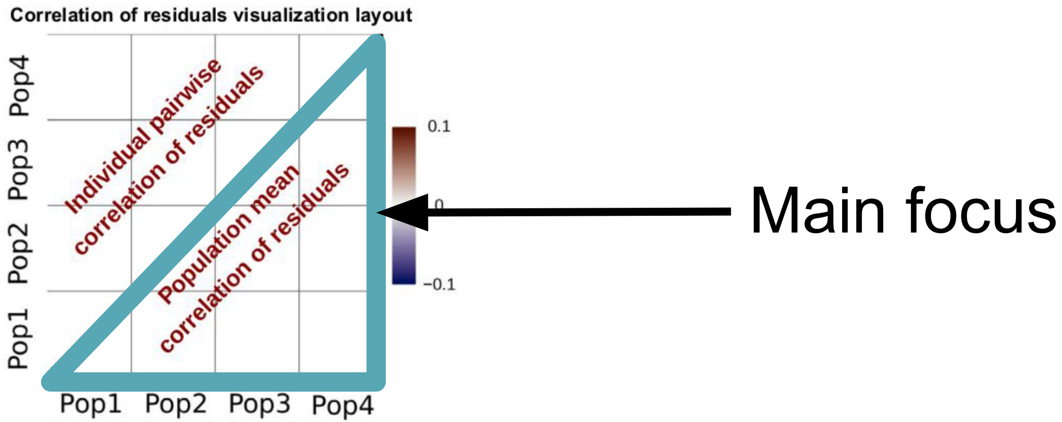
Potential solution to issues 2-4: evalAdmix

If individual *a* and *b* are sampled from the same population, and they have a good model fit, all error will be random and residuals will be uncorrelated:

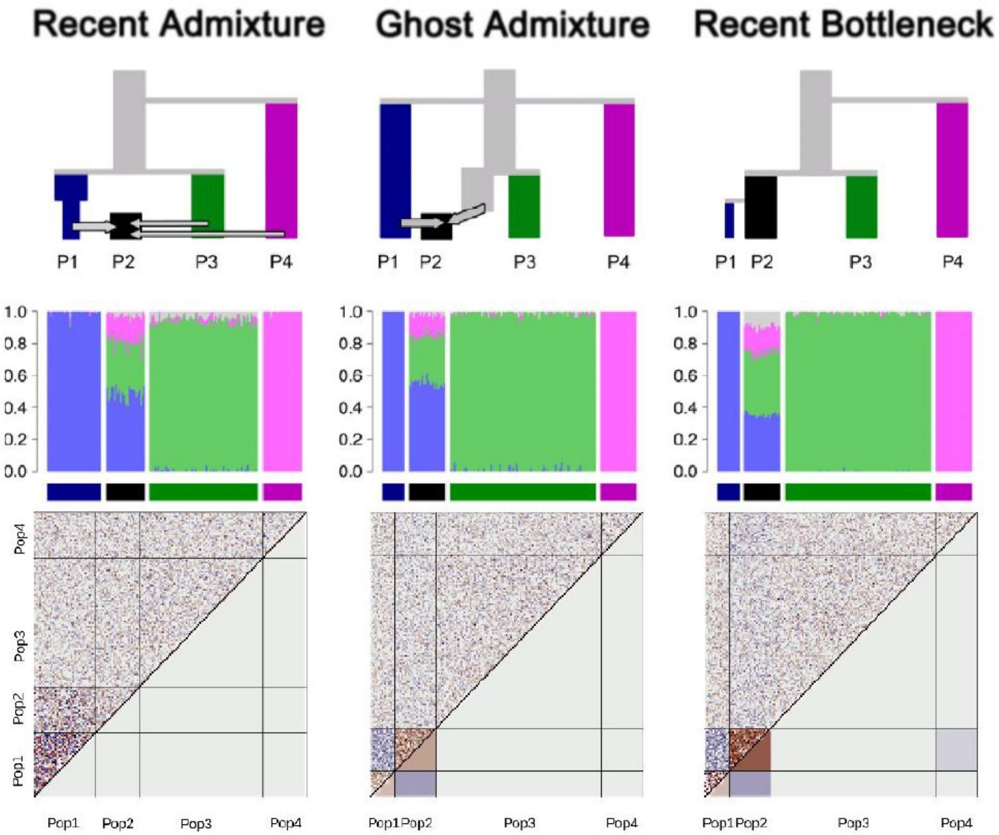
$$E[\hat{\rho}_{ab}] = 0$$

If individual *a* and *b* are sampled from the same population AND they have a bad model fit, they will share a systematic error and their residuals will be positively correlated:

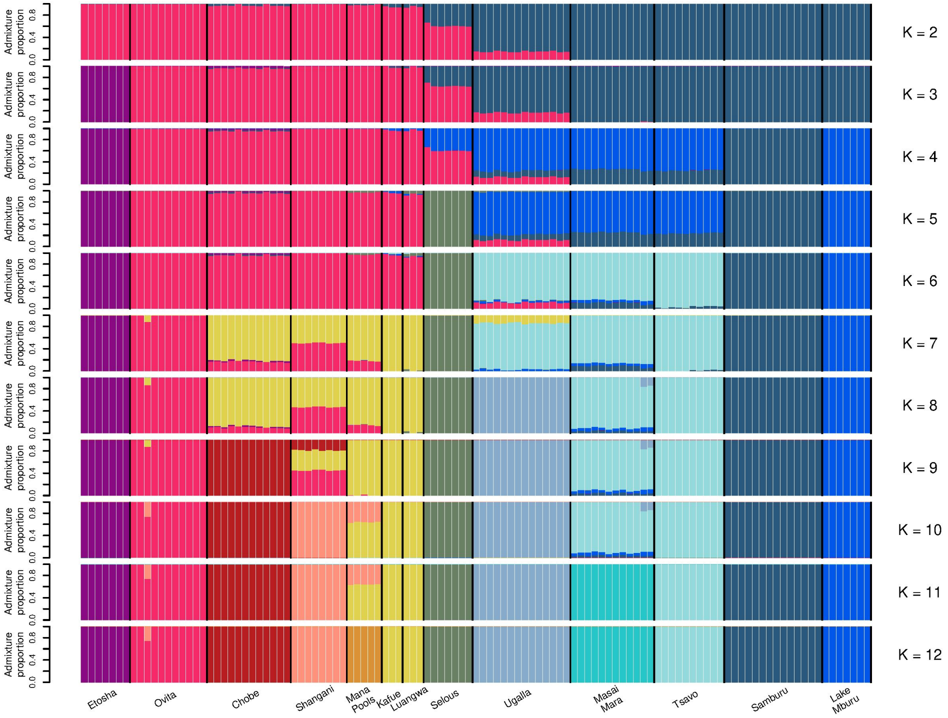
$$E[\hat{\rho}_{ab}] > 0$$

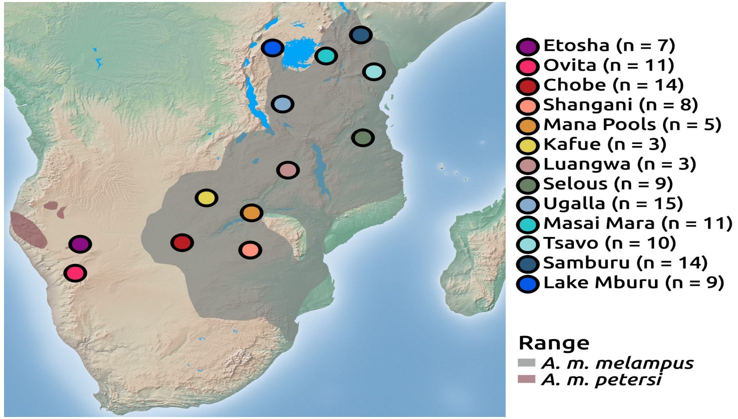


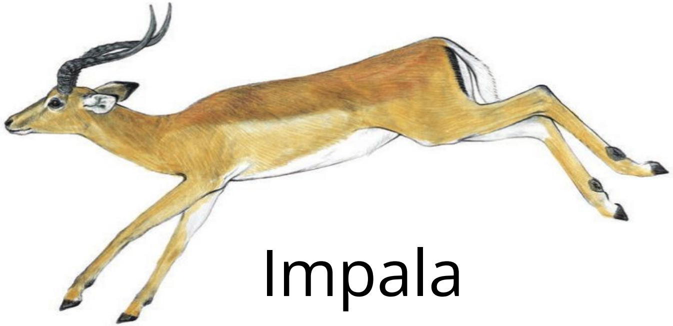
Potential solution to issues 2-4: evalAdmix



Example of evaluating the fit of admixture analyses

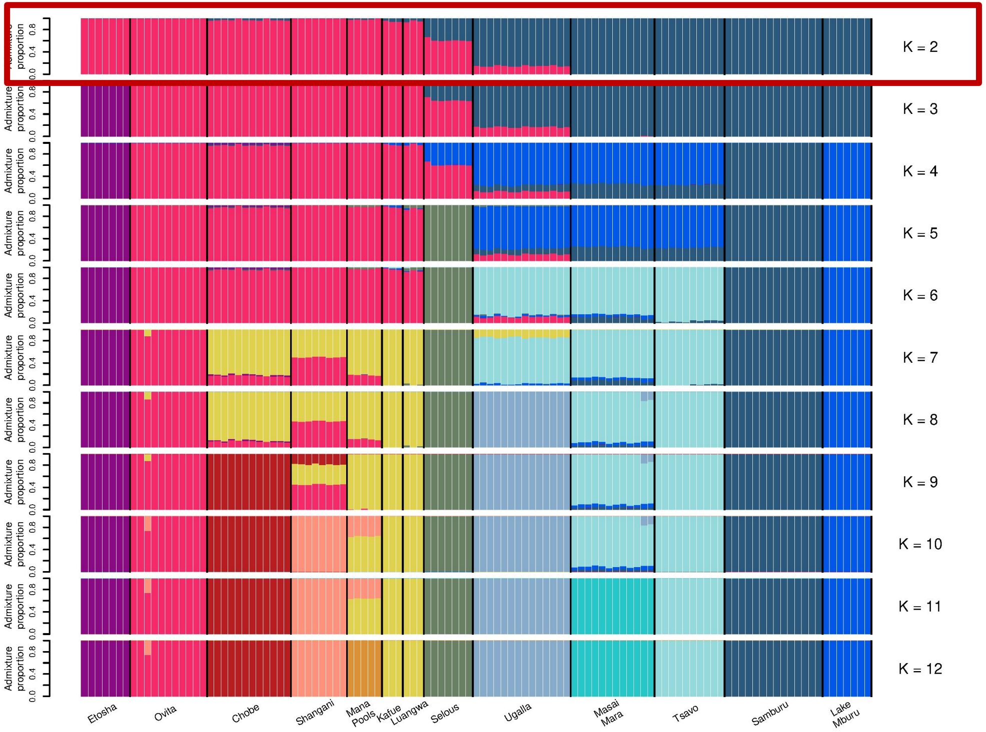


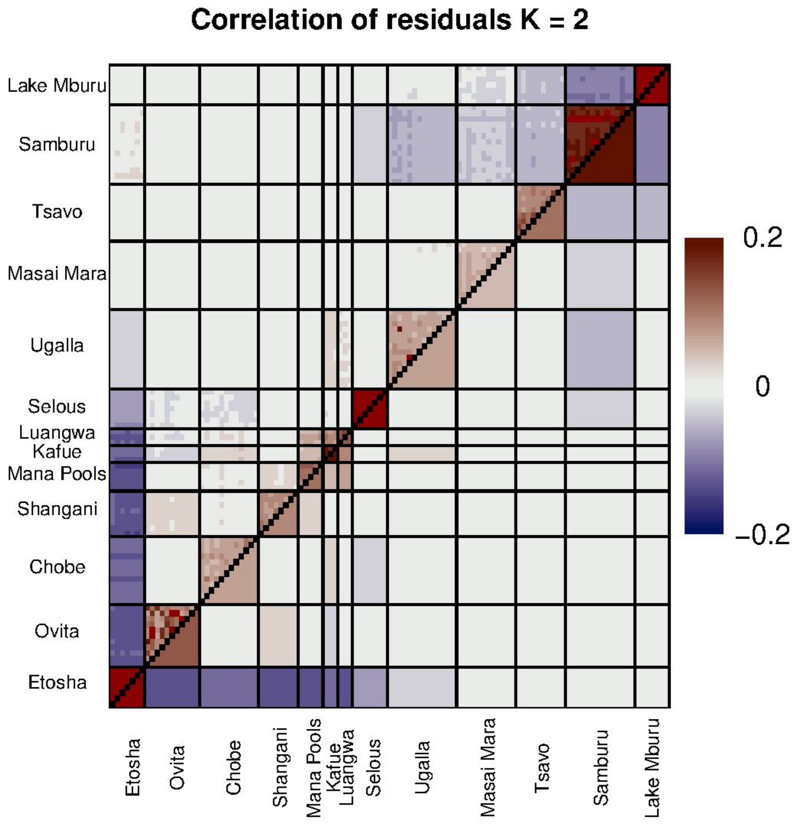




Impala

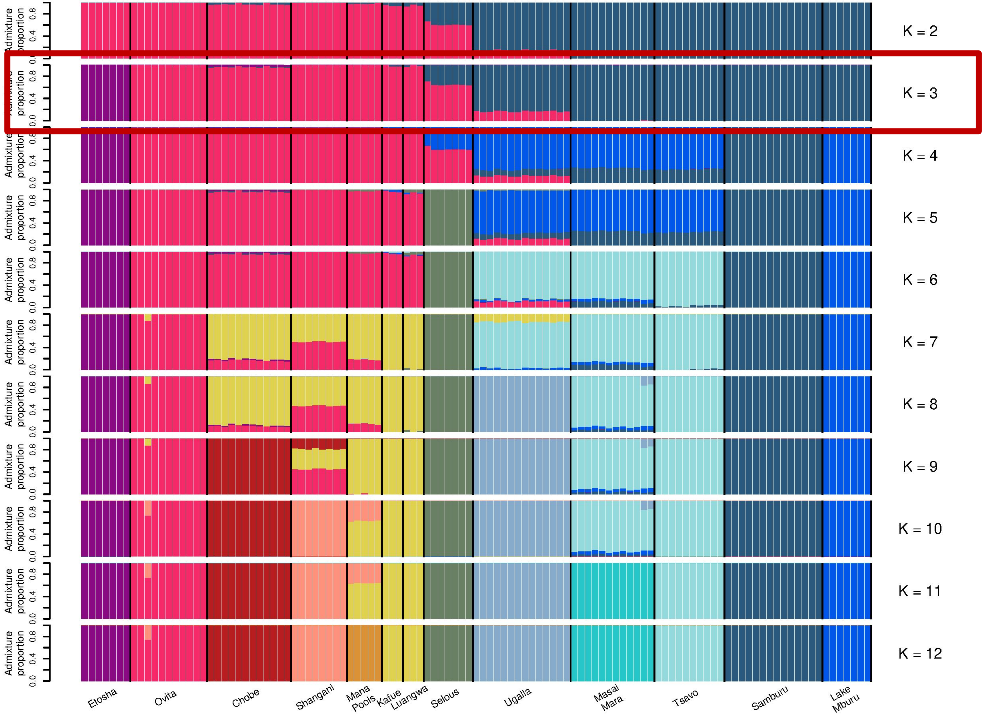
Example: impala again

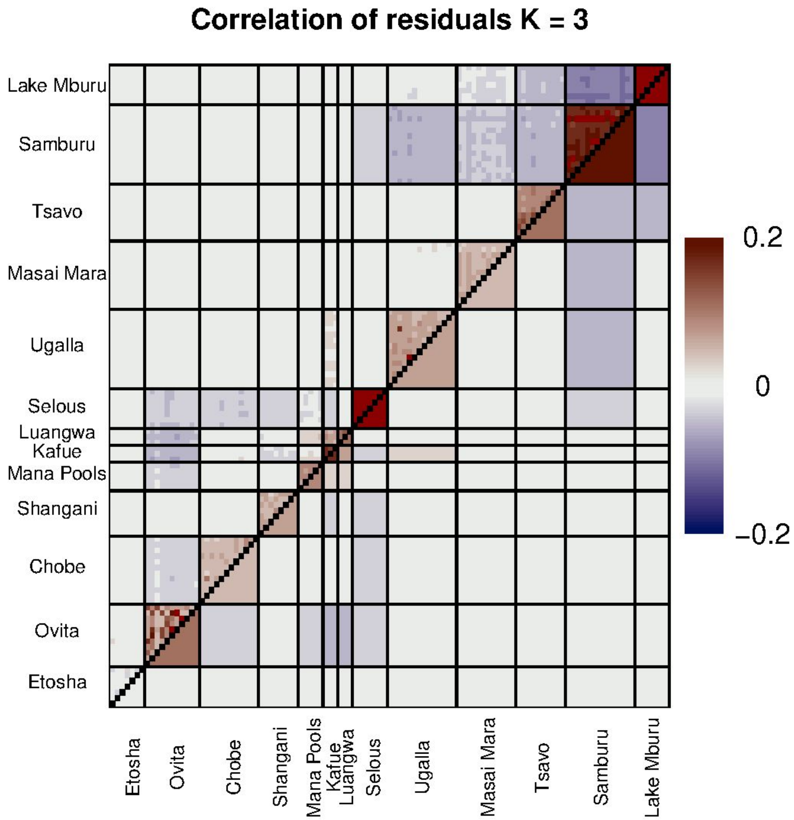




K=2

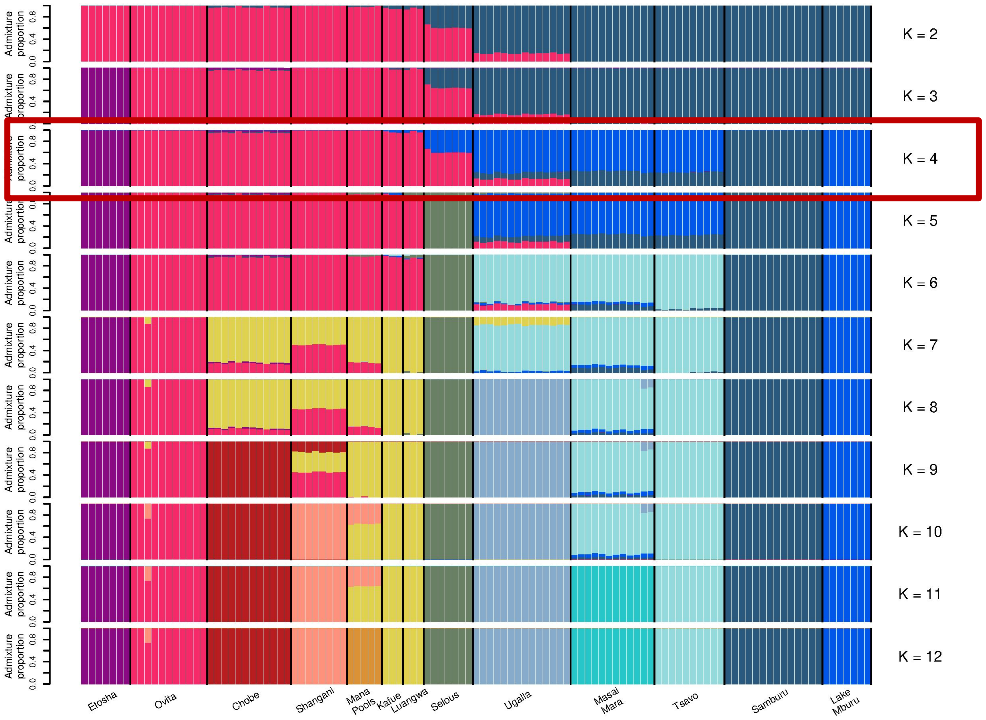
Example: impala again

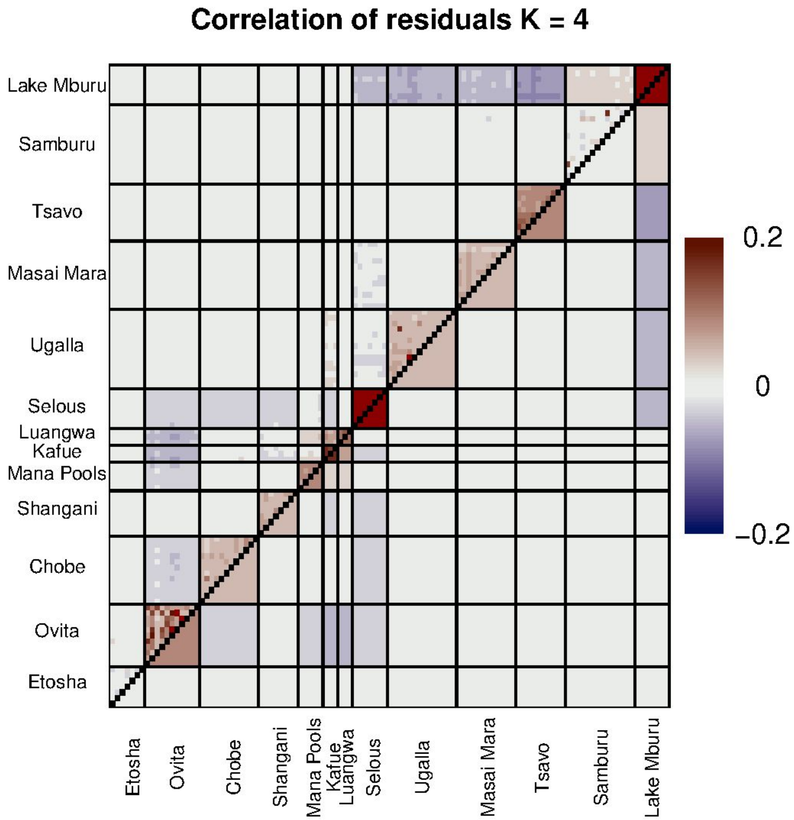




K=3

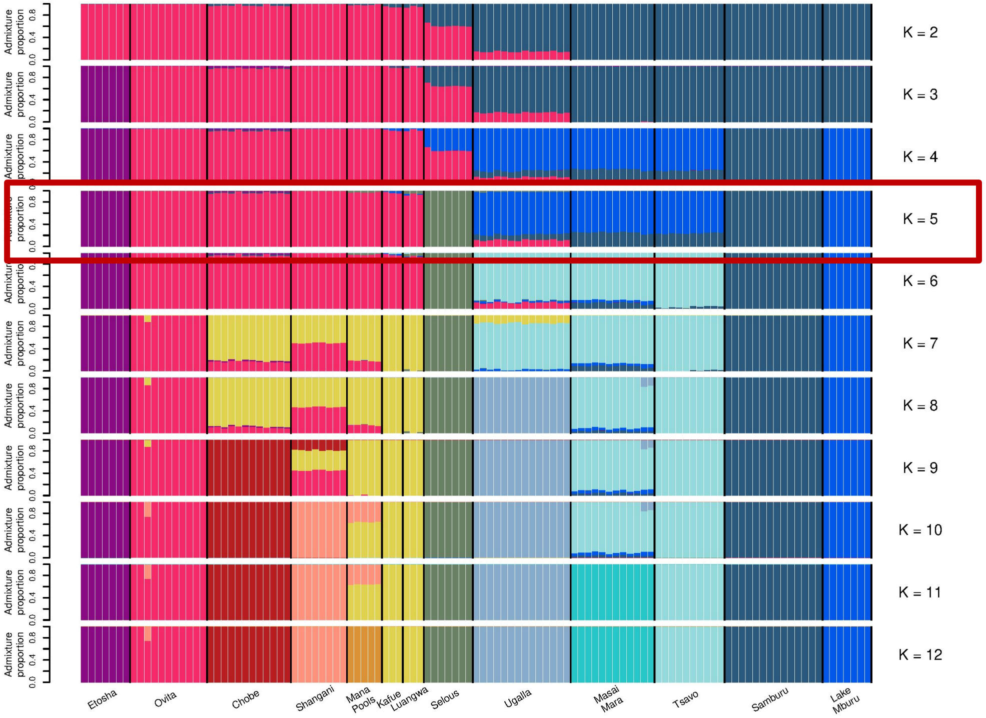
Example: impala again

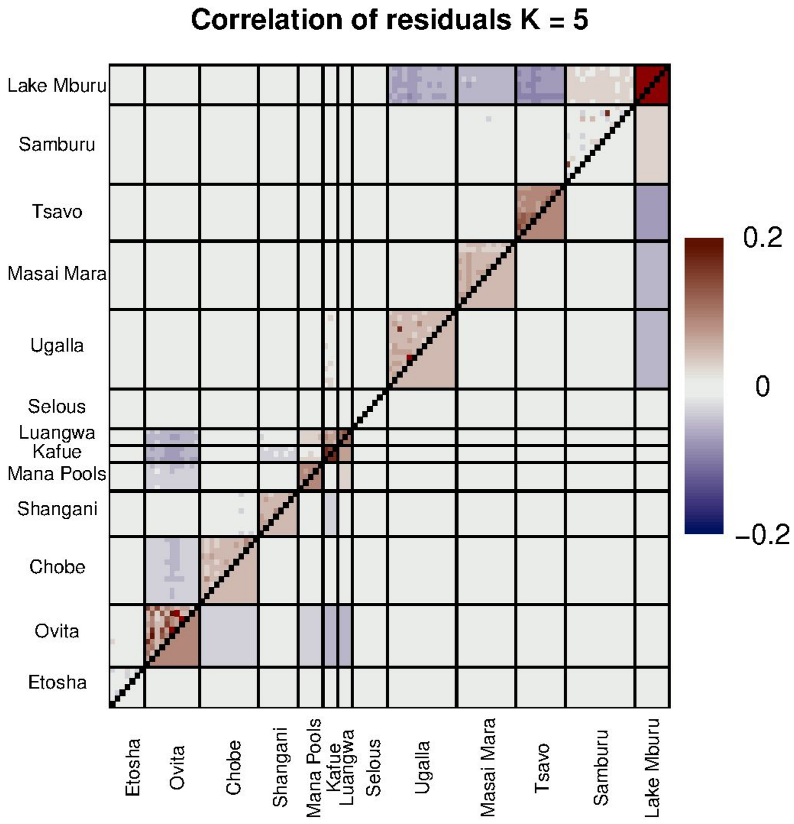




K=4

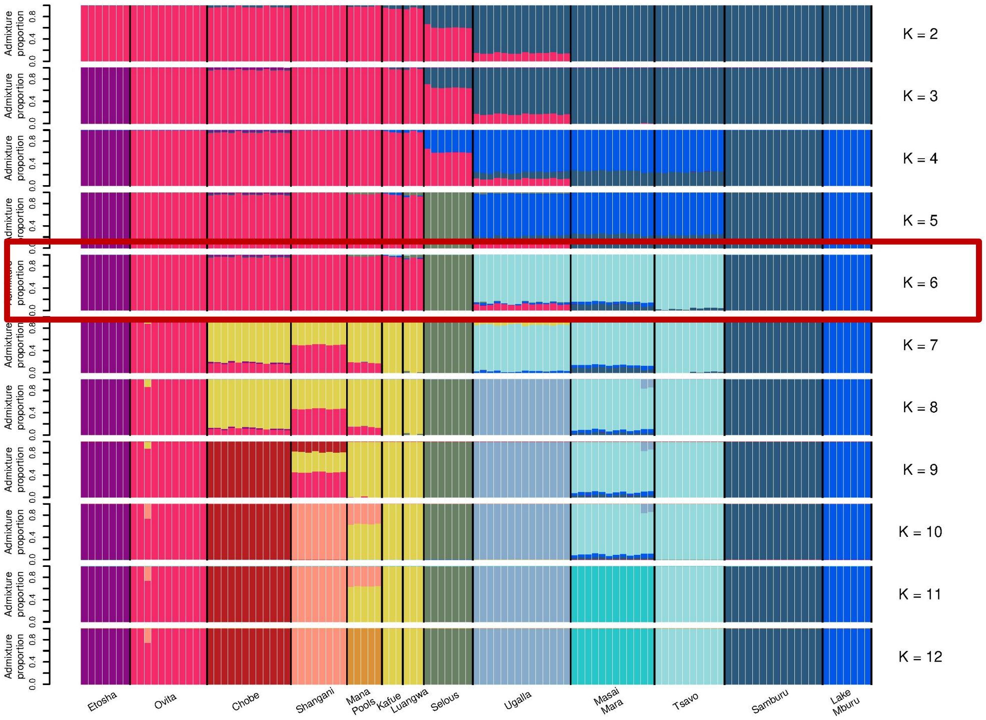
Example: impala again

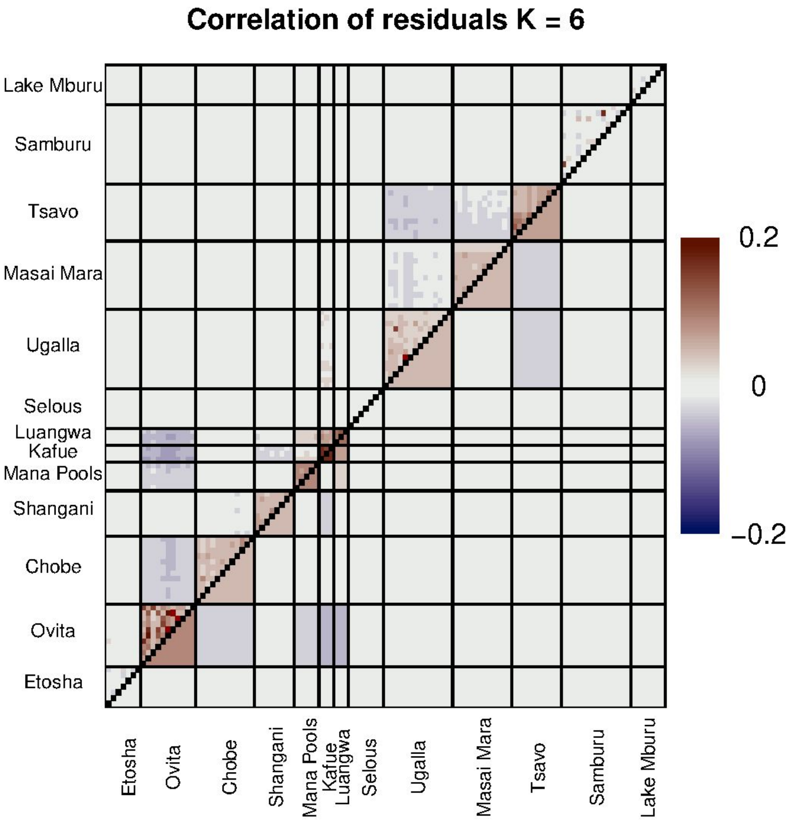




K=5

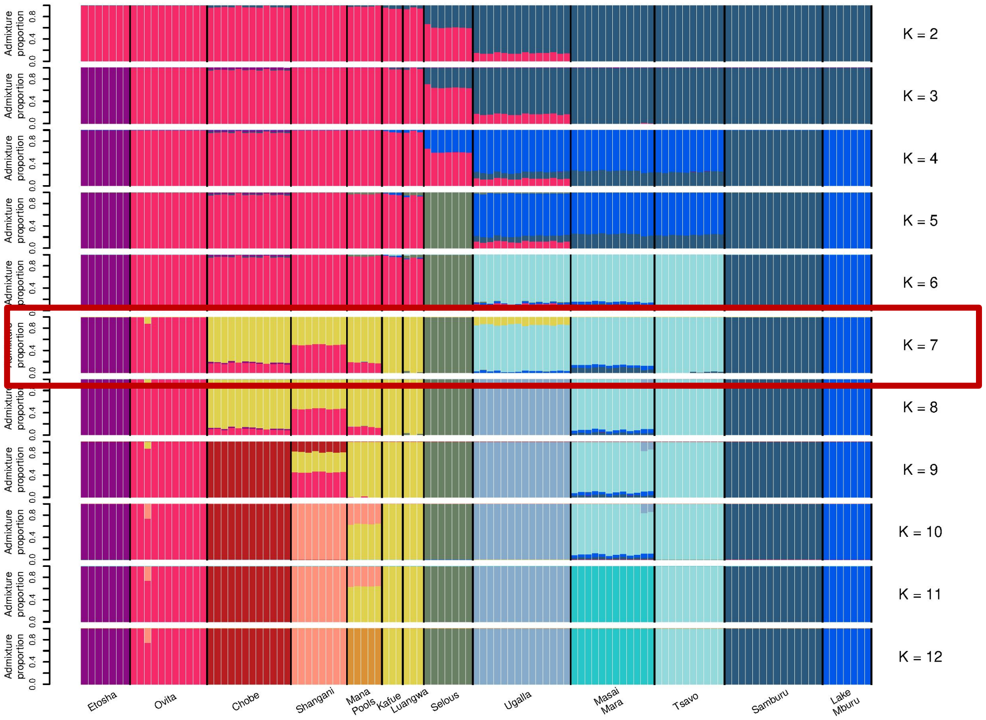
Example: impala again

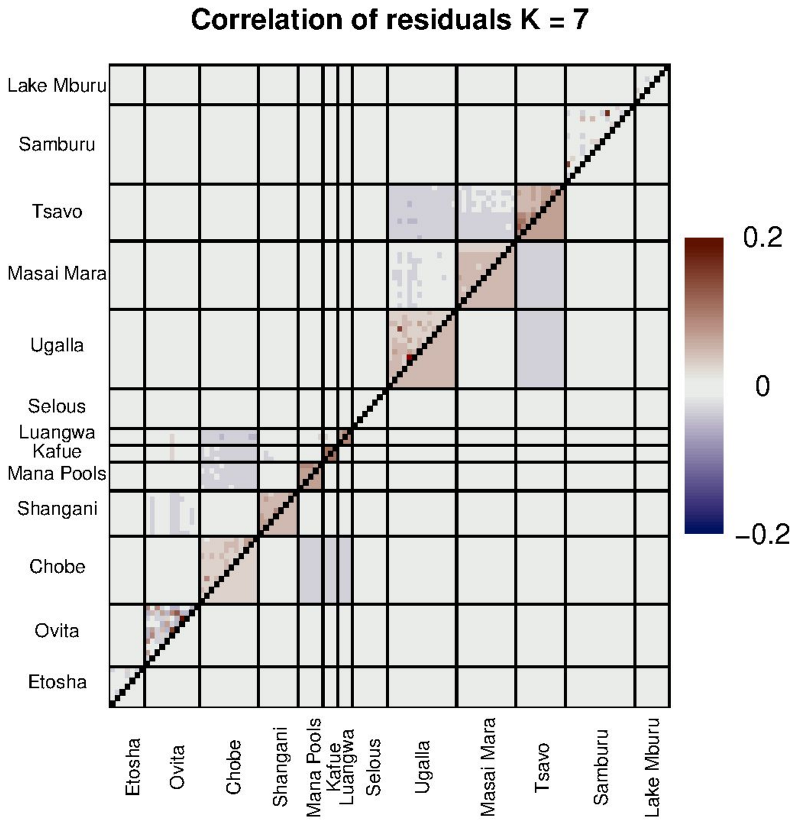




K=6

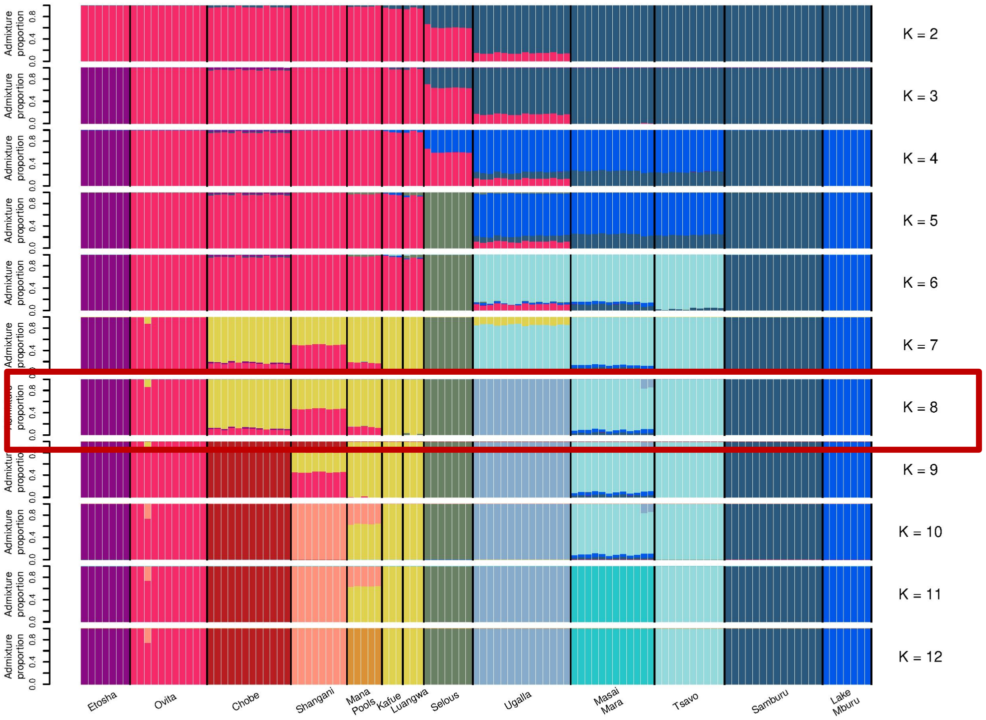
Example: impala again

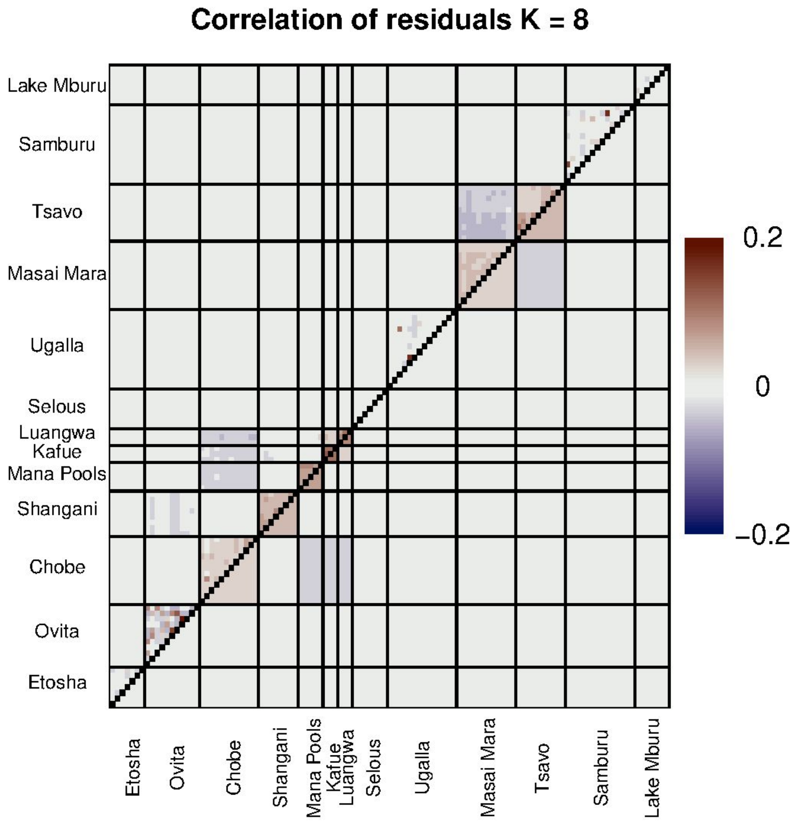




K=7

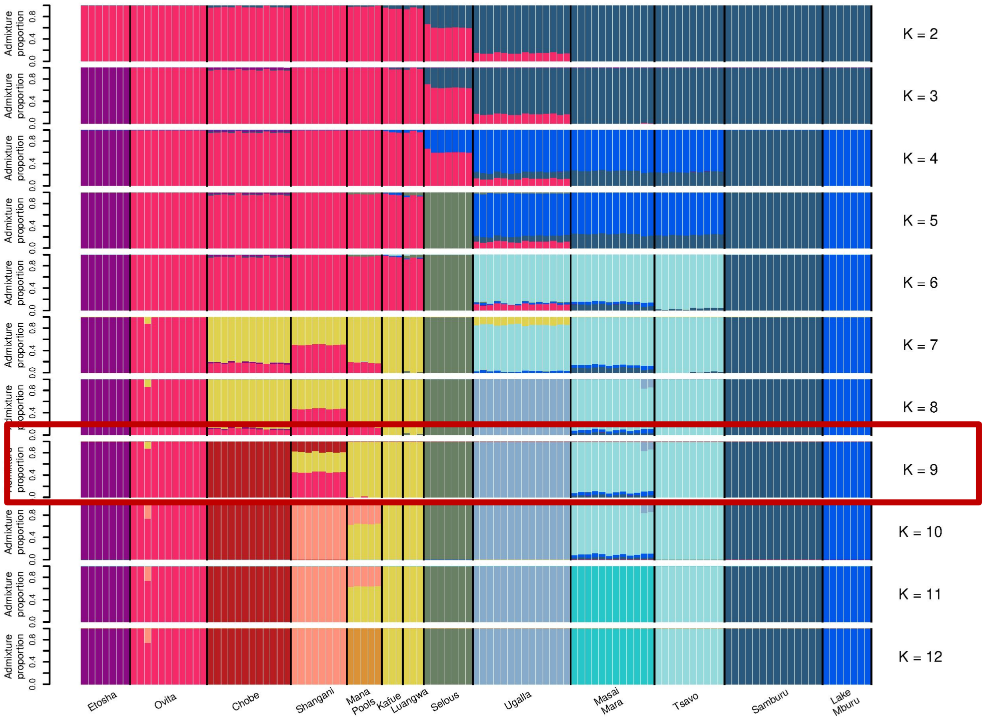
Example: impala again

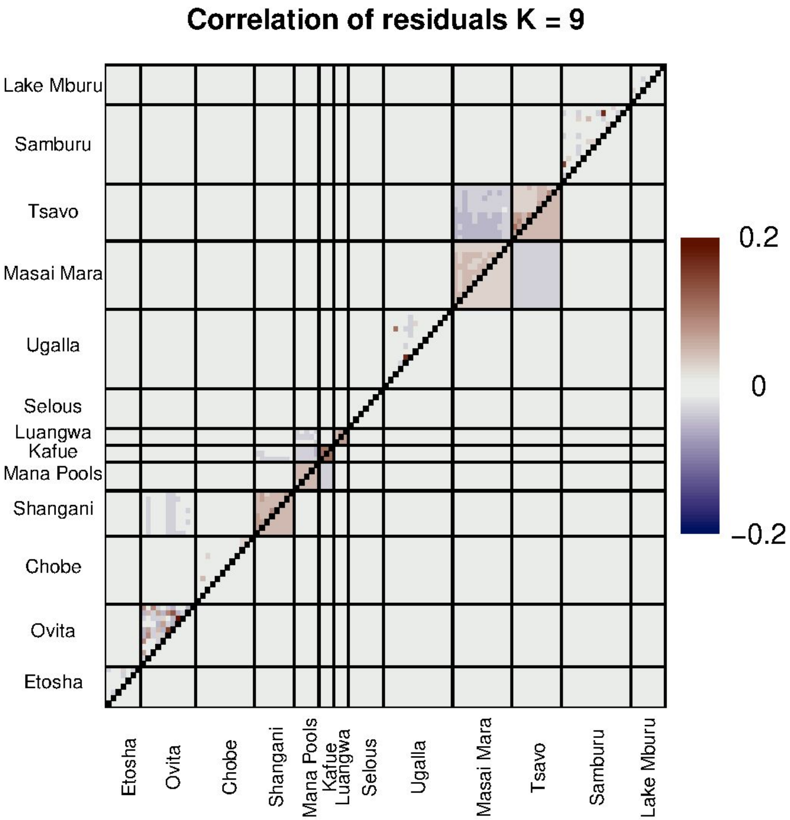




K=8

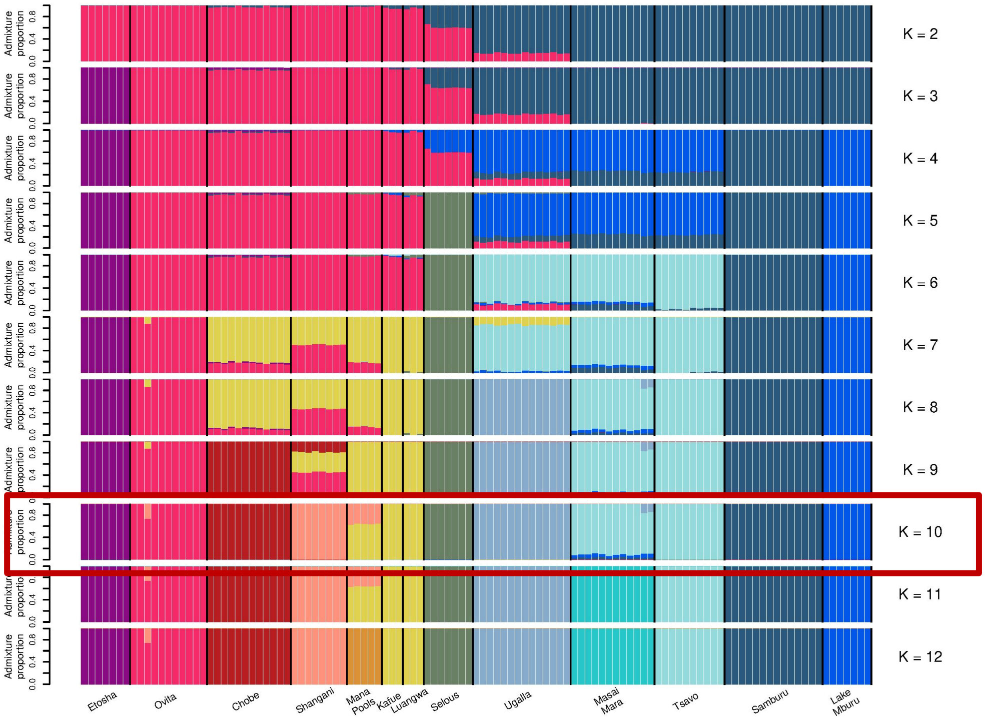
Example: impala again

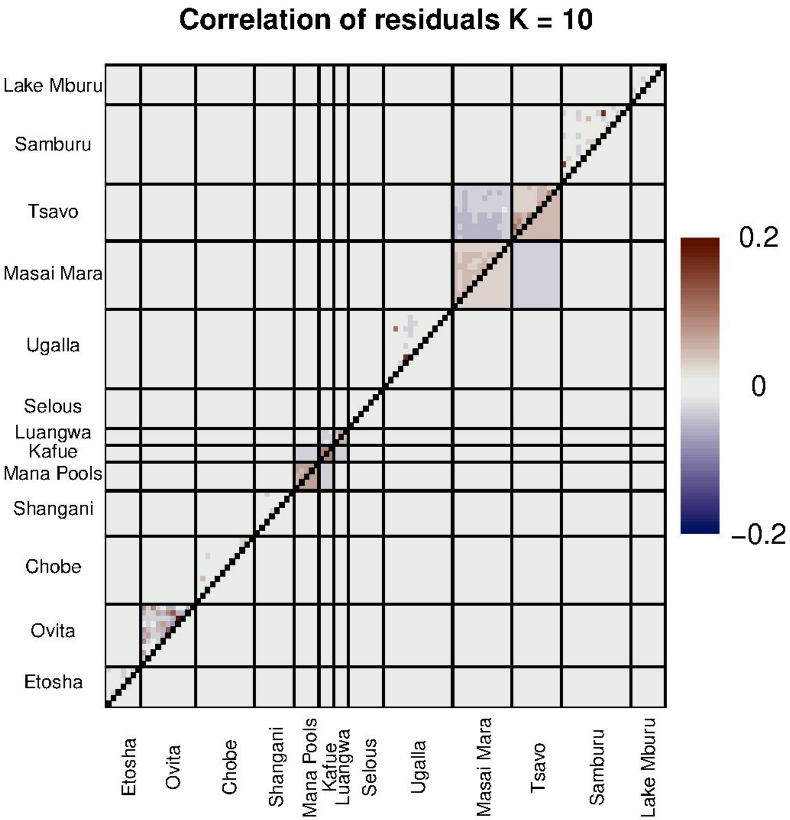




K=9

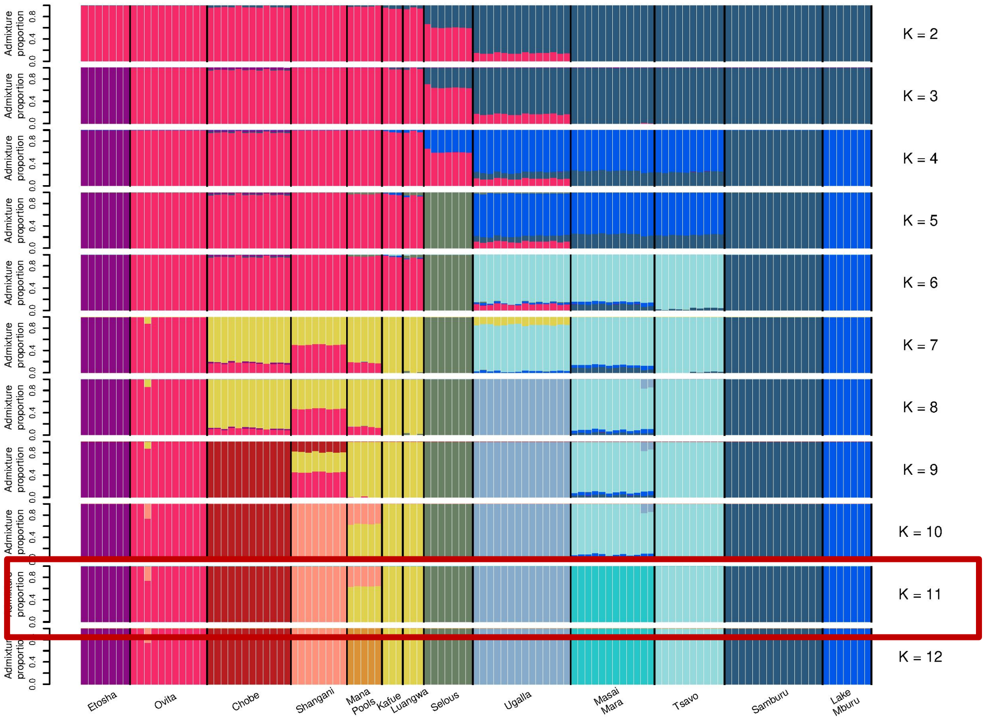
Example: impala again

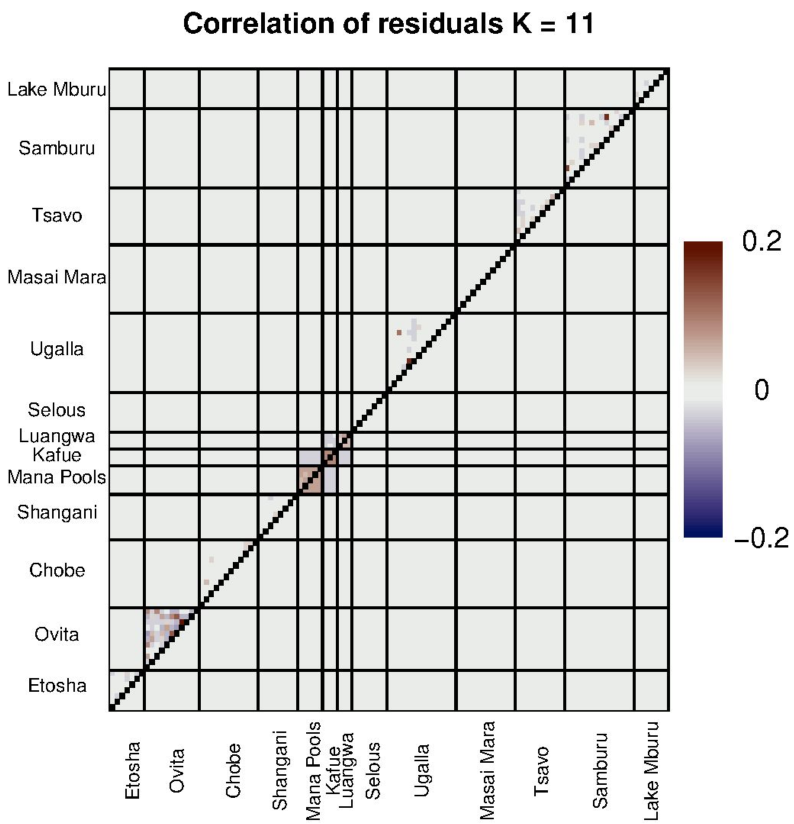




K=10

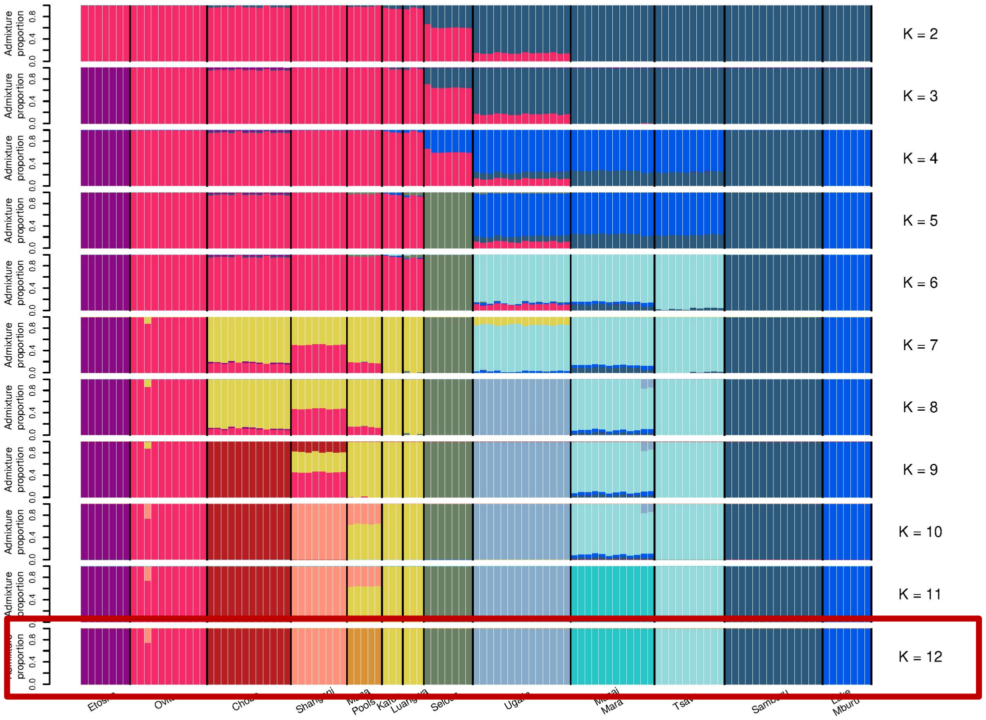
Example: impala again

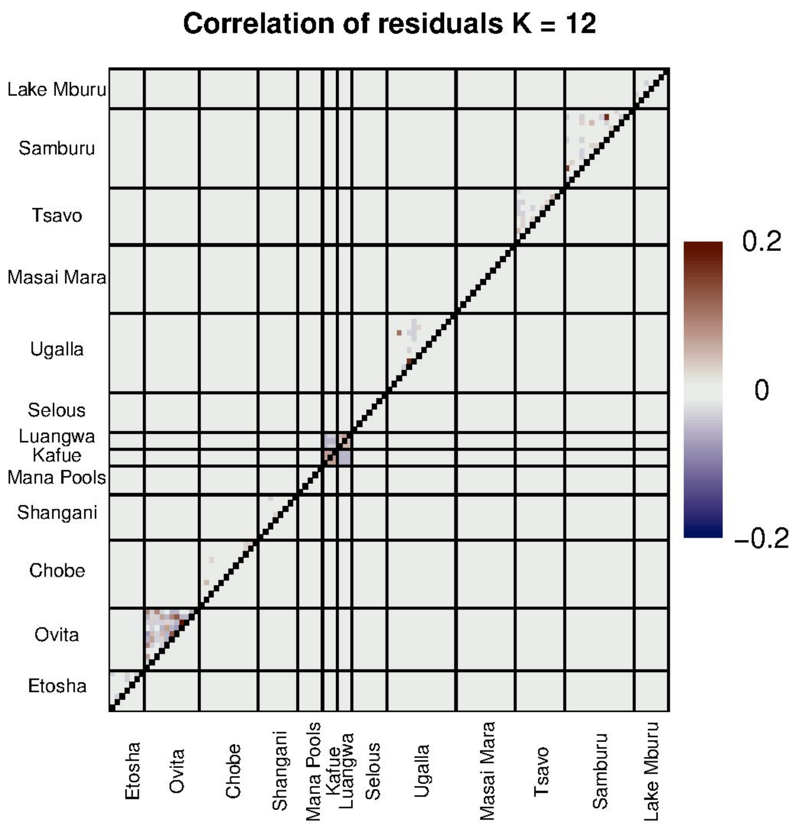




K=11

Example: impala again





K=12

More later :-)

- Be aware that ADMIXTURE only works for recent admixture
- For historical admixture/gene flow you need other methods
- Harvi will tell you more about such a method later :-)

Time for exercises

Run the admixture notebook

Extra slides

Solution: ADMIXTURE

- A model with ancestry-specific allele frequencies (F) and ancestry proportions (Q) as parameters and genotypes (G) **across all SNPs and all individuals** as the observed data
- To illustrate F, Q and G let's look a tiny example with 2 individuals and 8 SNPs where we for some reason know the ancestry (pop1 or pop2) of each allele:

G =

	SNP1	SNP2	SNP3	SNP4	SNP5	SNP6	SNP7	SNP8
Ind1	A	T	G	T	T	A	A	T
	T	T	G	C	T	G	T	T
Ind2	T	T	C	T	T	G	A	G
	T	G	C	T	A	G	T	T

Q =

	Pop1	Pop2
	4/16	12/16
	6/16	10/16

F =

Pop1	1/2	1/2	1/1	1/1	1/1	0/1	1/1	0/1
Pop2	0/2	2/2	1/3	2/3	2/3	1/3	1/3	3/3

G=

SNPs

IND 1	A	T	G	T	T	A	A	T
	T	T	G	C	T	G	T	T
IND 2	T	T	C	T	T	G	A	G
	T	G	C	T	A	G	T	T
…	T	G	G	T	T	G	A	G
	T	T	C	C	T	G	T	T
	A	T	G	T	T	A	A	T
	T	T	G	T	T	G	T	T
	T	T	G	T	A	G	A	G
	T	G	G	T	T	G	T	G

Q=

Pop 1	Pop 2
4/16	12/16
6/16	10/16
3/16	13/16
11/16	5/16
3/16	13/16

Admixture proportions

F=

Pop1	2/5	¾	4/4	3/3	3/3	0/3	2/3	0/2
pop2	0/5	4/6	3/6	5/7	5/7	2/7	3/7	6/8

Allele frequency

For one of i's alleles:

$$p(allele|Q^i, F^j) = q_{i1} f_{j1} + q_{i2} f_{j2} + \dots q_{ik} f_{jk} = h_{ij}$$

$$p(G_{ij}|Q_i, F_j) = \begin{cases} (h_{ij})^2 & \text{if } G_{ij} = 2, \\ 2h_{ij}(1 - h_{ij}) & \text{if } G_{ij} = 1, \\ (1 - h_{ij})^2 & \text{if } G_{ij} = 0. \end{cases}$$

$$p(G|Q, F) = \prod_i^N \prod_j^M p(G_{ij}|Q_i, F_j)$$

Q=

Pop 1	Pop 2
4/16	12/16
6/16	10/16
3/16	13/16
11/16	5/16
3/16	13/16

Admixture proportions

F=

Pop1	2/5	¾	4/4	3/3	3/3	0/3	2/3	0/2
pop2	0/5	4/6	3/6	5/7	5/7	2/7	3/7	6/8

Allele frequency

Example: wildebeest